
1755
МОСКОВСКИЙ
ГОСУДАРСТВЕННЫЙ
УНИВЕРСИТЕТ
ИМЕНИ М.В. ЛОМОНОСОВА
Visual Autoregressive
Models
• Predicts pixels iteratively
• Creates detailed and realistic
• Useful for generative AI

Image generation

A beautiful futuristic world with intelligent dinosaurs

Heroes of might and magic V; inferno vs necropolis fight; 100 weeks

GPT Image 1.5

Image editing

GPT Image 1.5

Video generation

Animate the people in these images so they come to life and step out of the photos...

Preserve their facial features, expressions, and identity as accurately as possible.

Transform each person into a different animal. After the transformation, place them inside a luxurious Bugatti Veyron.

Show them seated naturally as if on a road trip. Then create a cinematic sequence of the car driving upward, leaving Earth and traveling through space toward the Moon, with dramatic lighting, stars, and planetary scenery in the background.

KlingAI 2.6

World generation

A vibrant 3D style, an adorable, fluffy creature bounding across a vibrant rainbow bridge in a fantastical landscape.

The rainbow bridge arches gracefully through a whimsical landscape, perhaps filled with floating islands, glowing flora, and swirling clouds.

...

Genie 3

Useful materials

Visual GenAI at YSDA

Basic GM courses

Stanford CS236 Deep Generative Models

Deep Generative Models at YSDA by Roma Isachenko

Diffusion models at HSE by Denis Rakitin

Visual GenAI Research Seminars

RU GenAI research community

Weekly seminars and paper reviews

Stanford CS236: https://www.youtube.com/watch?v=XZ0PMRWXBEU
DGM at YSDA: https://github.com/r-isachenko/2025-DGM-MIPT-YSDA-course
The Principles of Diffusion Models: https://the-principles-of-diffusion-models.github.io/

Diffusion-based models is a leading paradigm for visual generation

Diffusion

AR models

Why AR visual models?

Making use of pretrained LLMs: even text-text pretrain is useful for visual generation

Why AR visual models?

Making use of pretrained LLMs: even text-text pretrain is useful for visual generation

Not AR vs Diffusion, but AR + Diffusion

e.g.

Long video generation: diffusion for spatial dimensions and AR for temporal

Outline

Goal for today: understanding visual AR basics

01 **Classical visual AR modeling**

02 **Next-scale prediction modeling**

Image tokenization

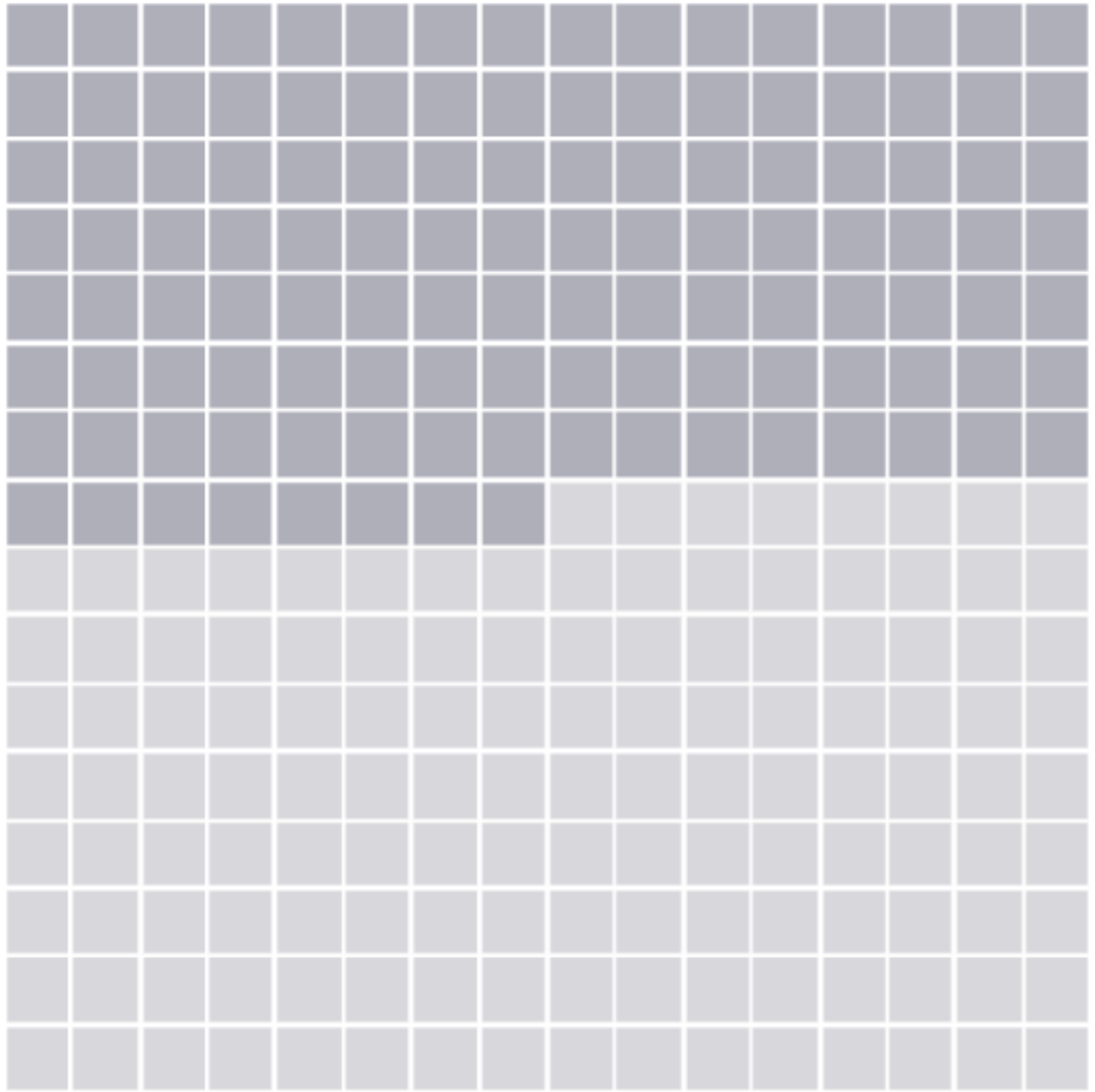
Similar to LLMs, we first need to represent **images as a sequence of discrete tokens**

Naive image tokenization

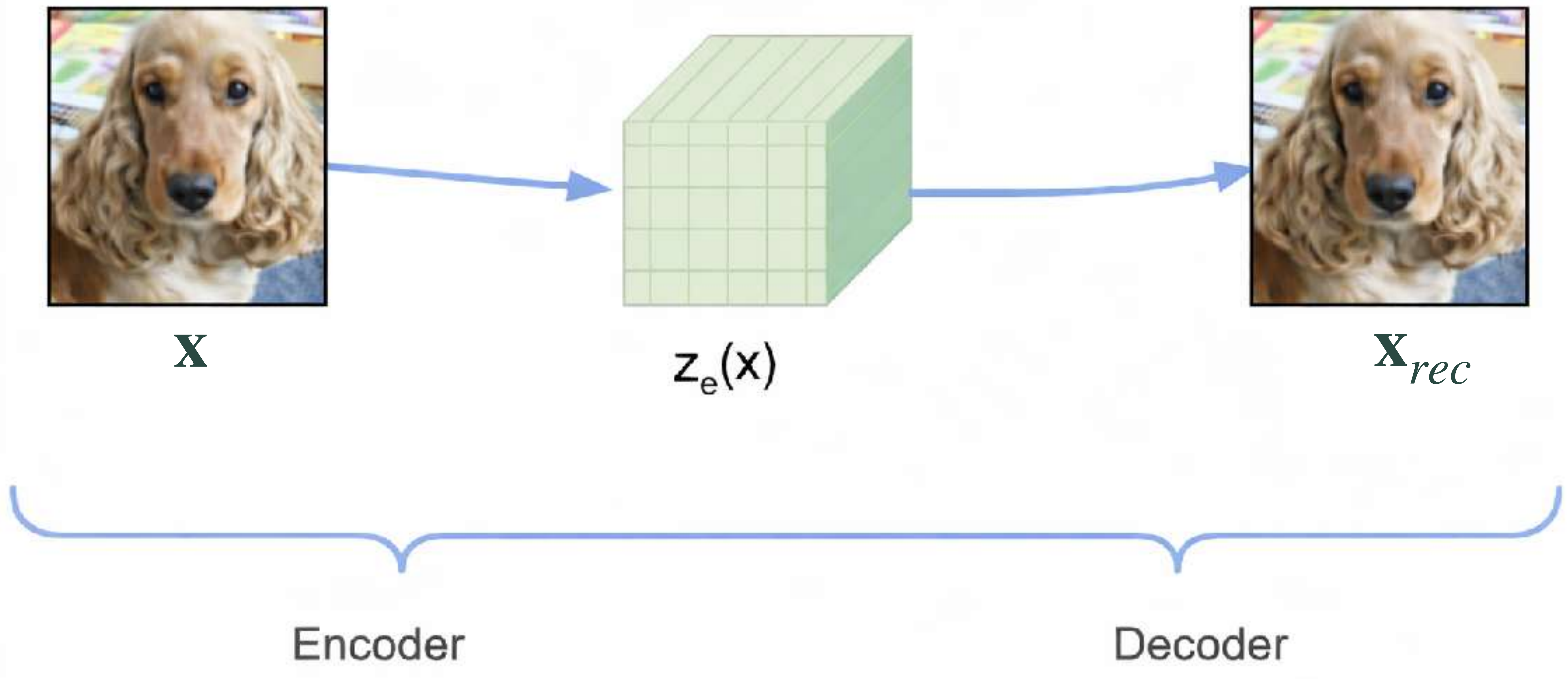
Pixels: each pixels is a discrete value in [0, 255]

Extremely expensive even for low-resolution images

Raster-order image generation

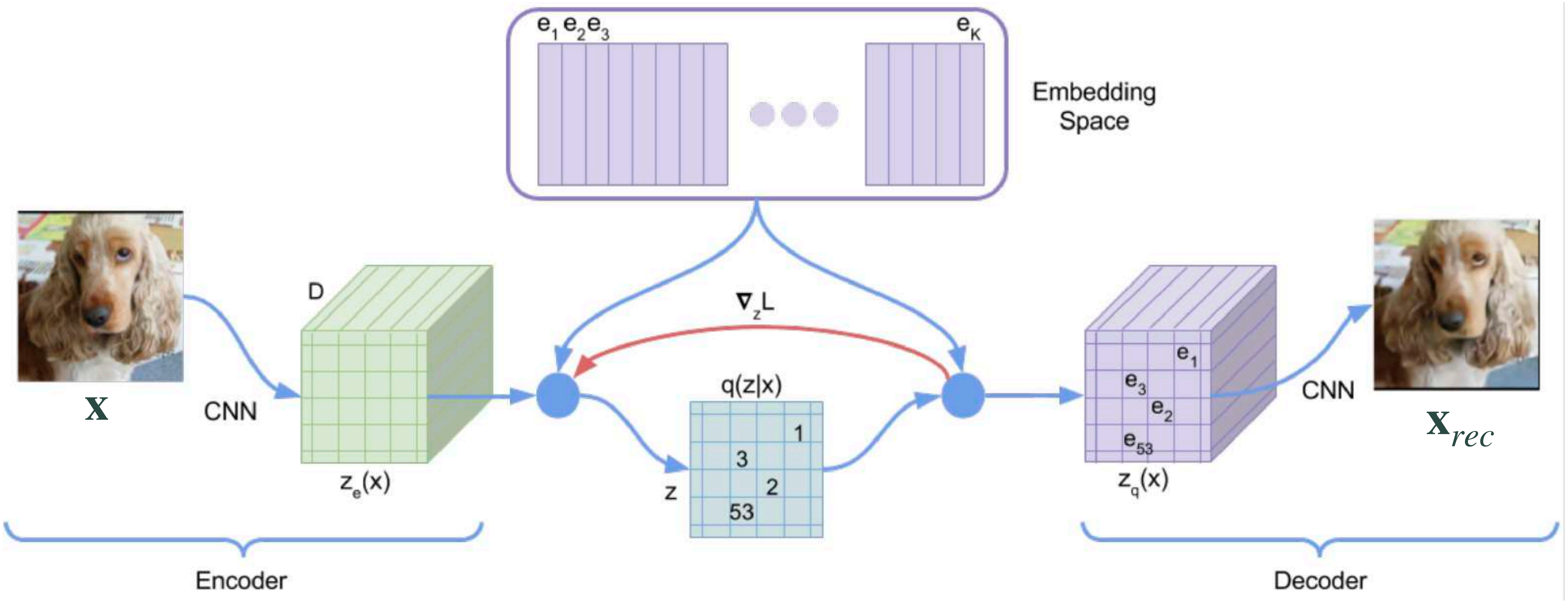


Continuous Autoencoders

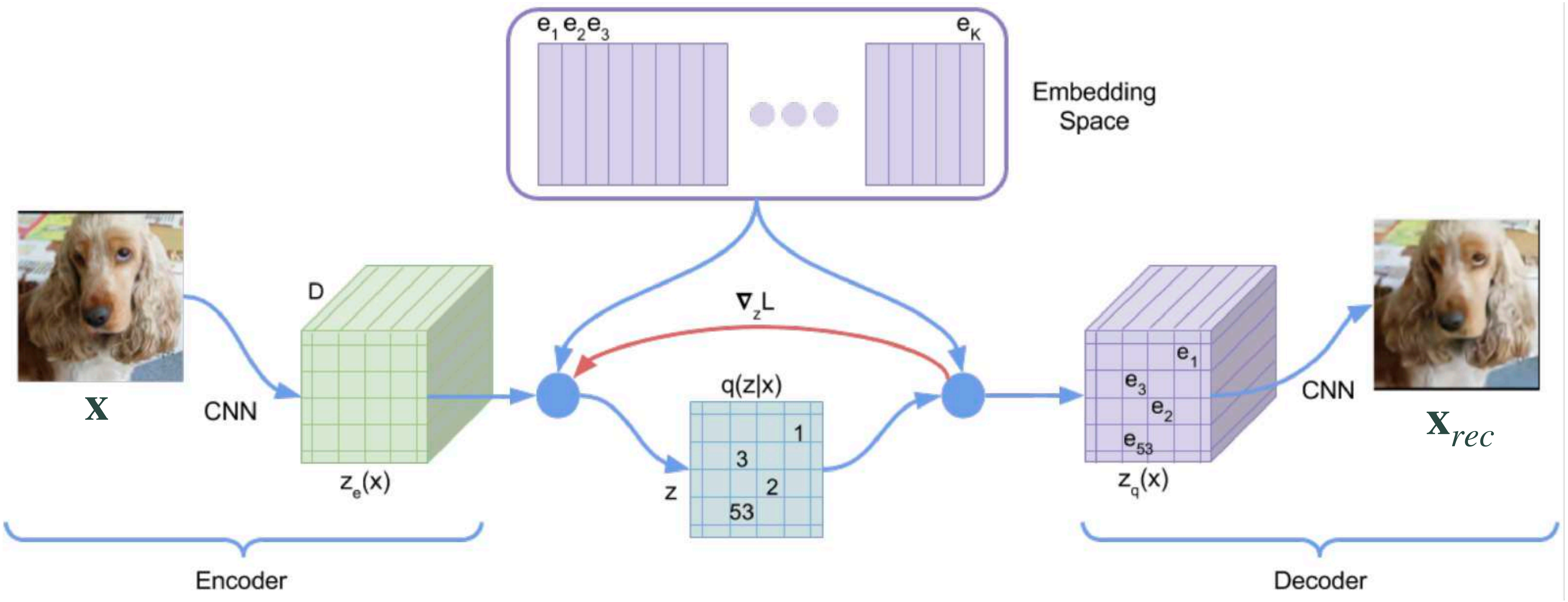


Reconstruction loss: $L = \|\mathbf{x} - \mathbf{x}_{rec}\|_2^2$

Vector Quantization Autoencoders



Vector Quantization Autoencoders



VQ-VAE loss: $L = \|\mathbf{x} - \mathbf{x}_{rec}\|_2^2 + \|\text{sg}[\mathbf{z}_e(x)] - e\|_2^2 + \beta\|\mathbf{z}_e(x) - \text{sg}[e]\|_2^2$

VQ-VAE follow-ups

VQ-GAN — VQ-VAE but add GAN and LPIPS losses for better reconstruction

ViT-VQ-GAN

— Convolutional AE → ViT-AE (Not critical)

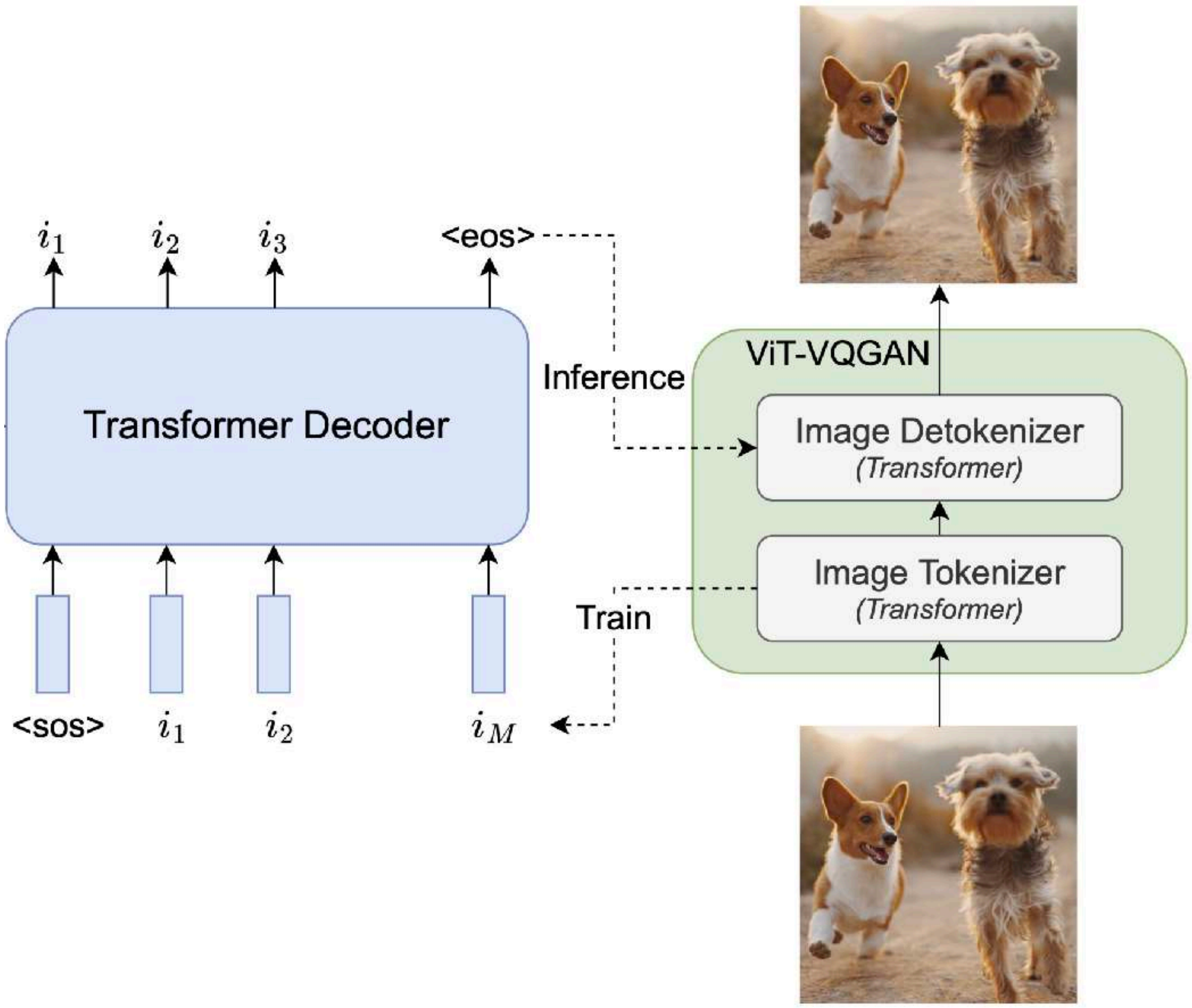
Address bad codebook utilization:

— Normalize latents and codebook vectors

— Reduce dimensionality of codebook vectors

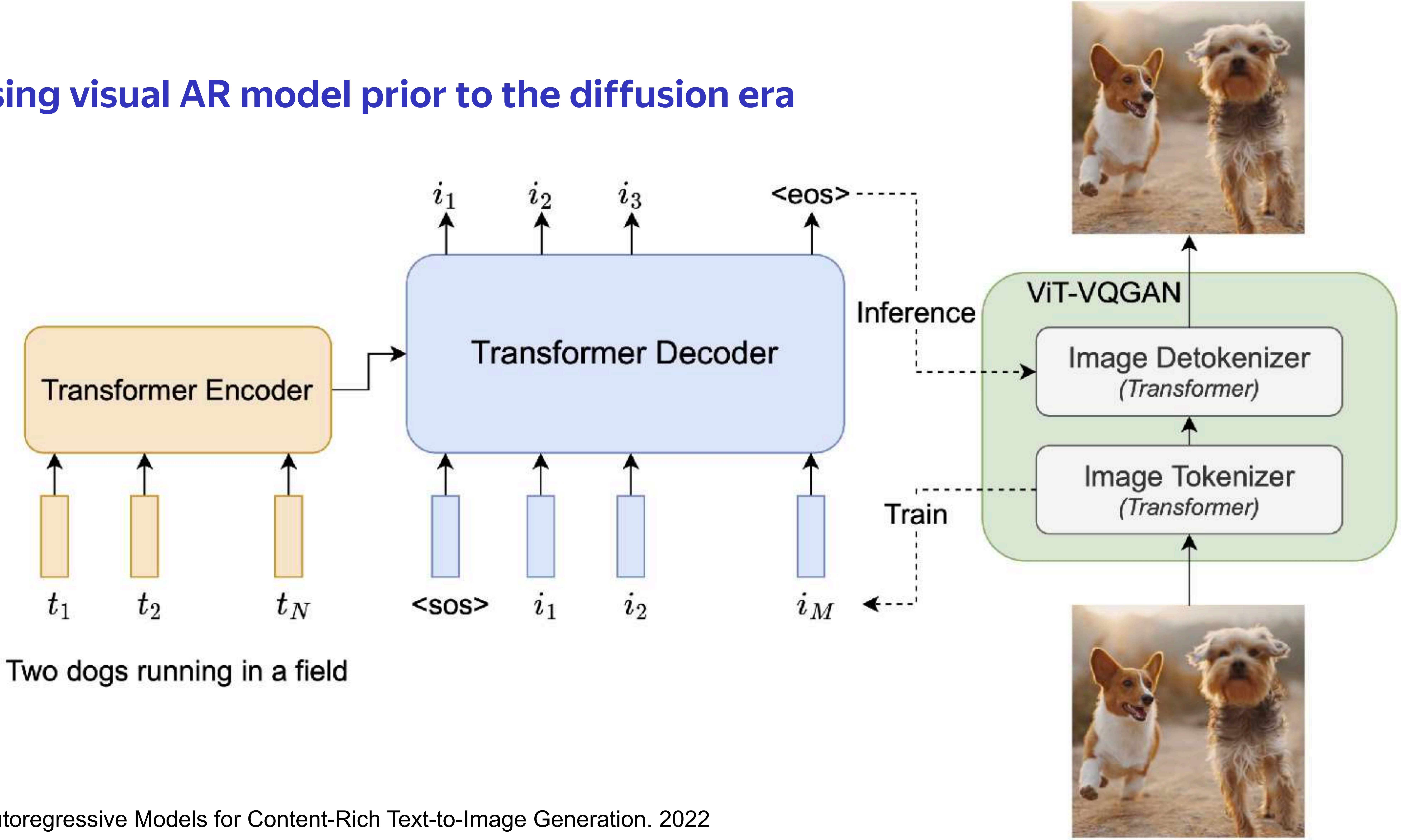
In practice: convolutional VQ-GAN + ViT-VQ-GAN tricks

Image AR generation



Text-to-image AR generation | Parti

Last promising visual AR model prior to the diffusion era



Yu et al., Scaling Autoregressive Models for Content-Rich Text-to-Image Generation. 2022

Sampling with discrete models

All typical sampling techniques from LLMs are valid here

Top-k and Top-p (nucleus) sampling

Temperature in softmax

Beam search *but I didn't see it in practice*

There are more image specific techniques

CFG: $\text{logits}_{cfg} = \text{logits}_{uncond} + \gamma \cdot (\text{logits}_{cond} - \text{logits}_{uncond})$

Gumbel-softmax averaging *quite specific thing, never mind*


A FEW
~~MOMENTS~~ LATER
YEARS

LlamaGen

Using more recent LLM architectures

RMSNorm pre-normalization

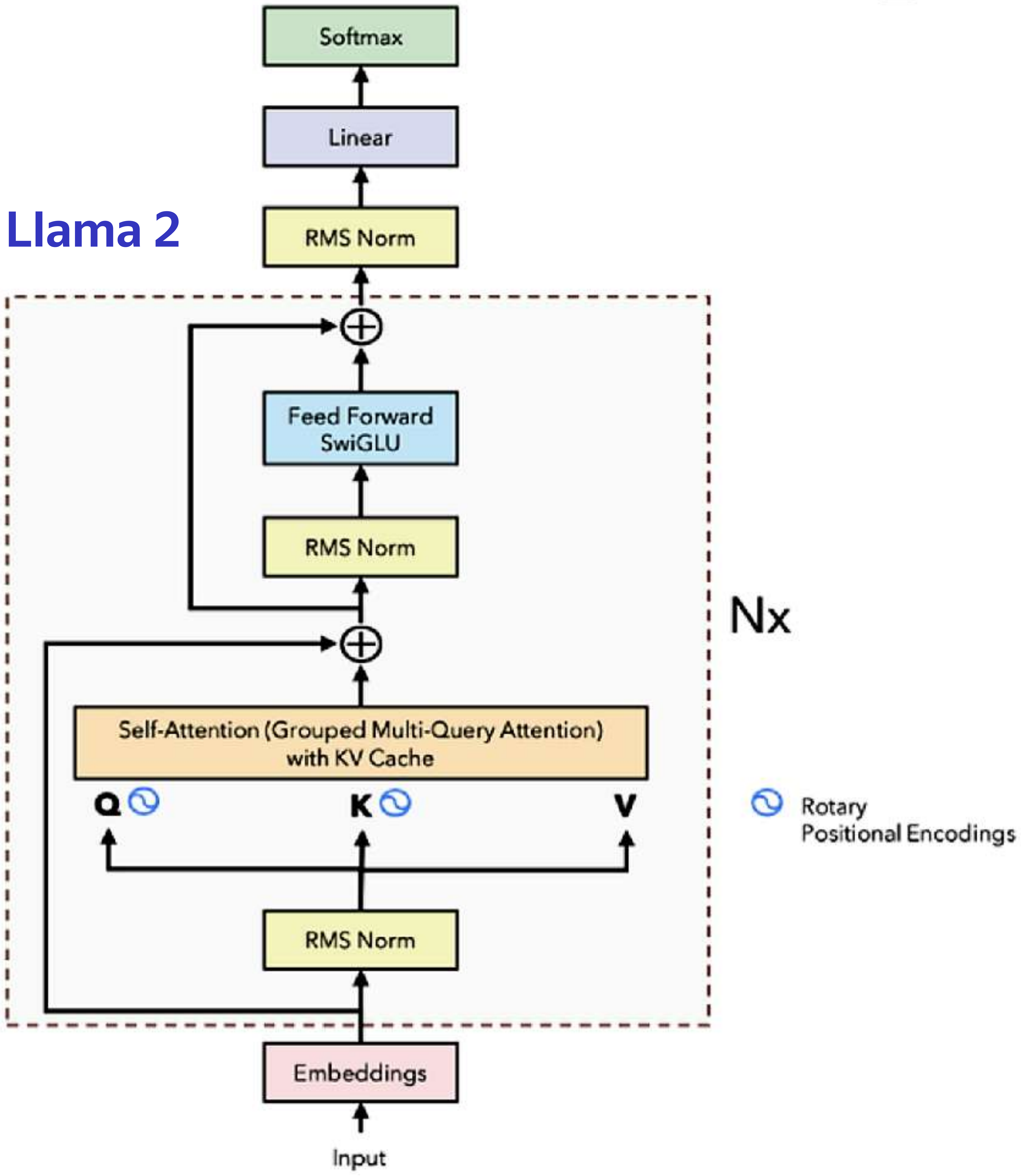
2D RoPE

In-context conditioning

SwiGLU

Moving closer to diffusion but no...

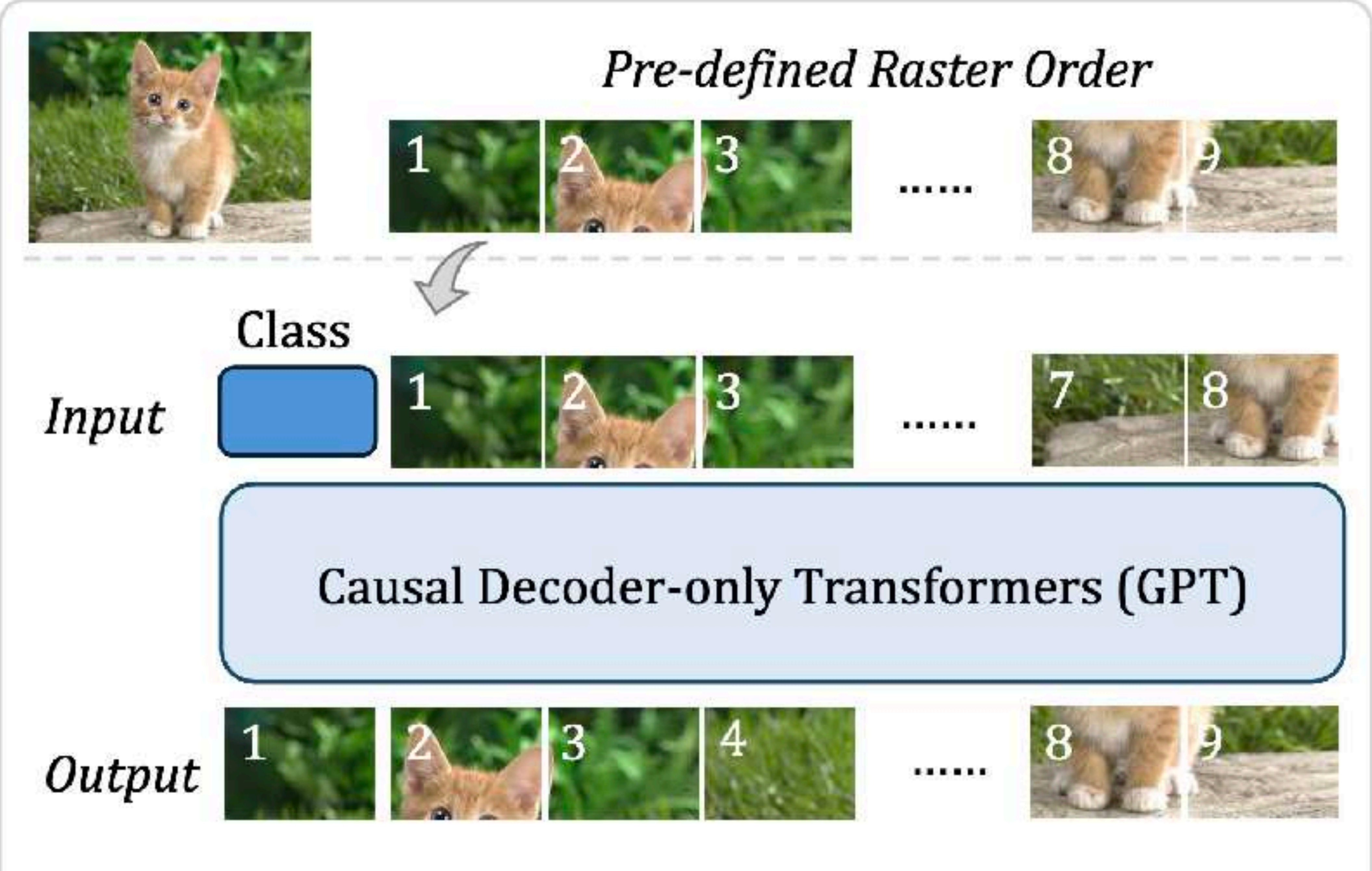
Llama 2



Sun et al., Autoregressive Model Beats Diffusion: Llama for Scalable Image Generation. 2024

Raster-order AR models

LlamaGen performs raster-order AR generation

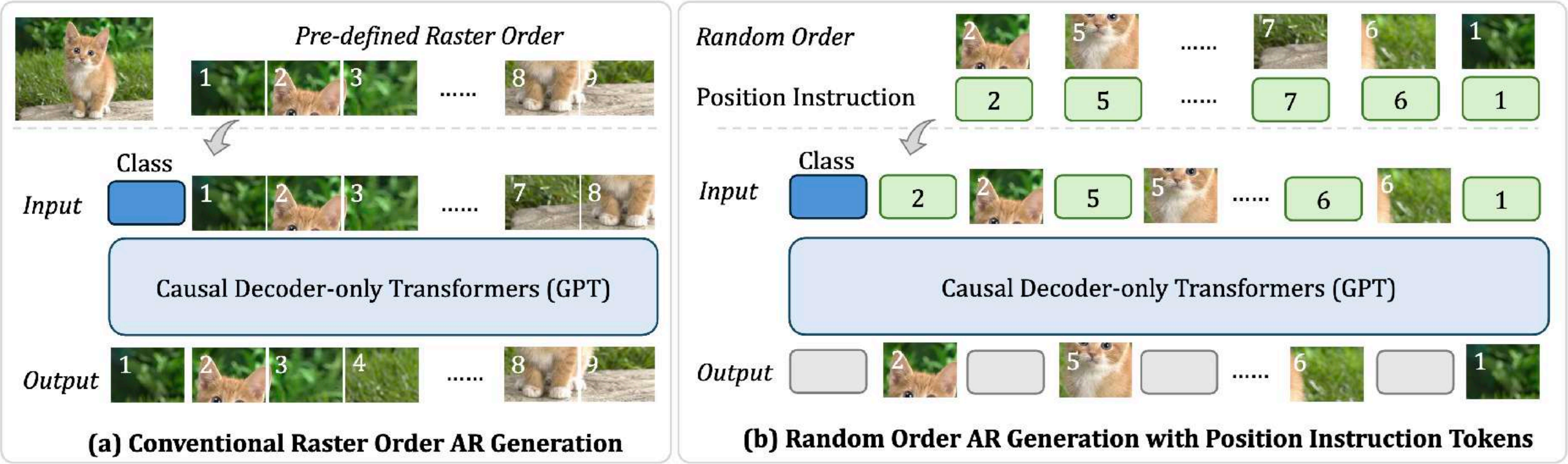


(a) Conventional Raster Order AR Generation

Pang et al., RandAR: Decoder-only Autoregressive Visual Generation in Random Orders. 2024

Any-order AR models

RandAR enables AR generation in any orders by introducing Position Instruction Tokens



Pang et al., RandAR: Decoder-only Autoregressive Visual Generation in Random Orders. 2024

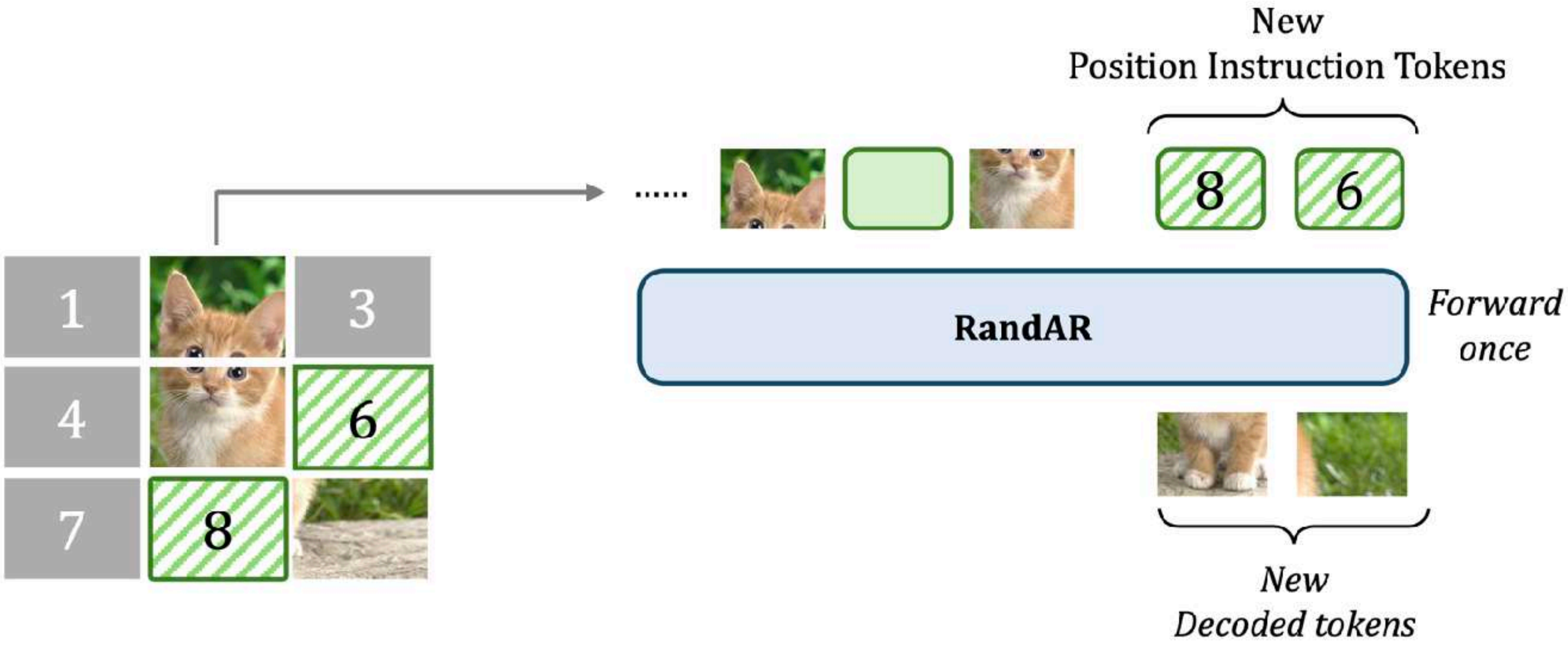
Any-order AR models

Does any order improve image quality? No, but...

Pang et al., RandAR: Decoder-only Autoregressive Visual Generation in Random Orders. 2024

Any-order AR models

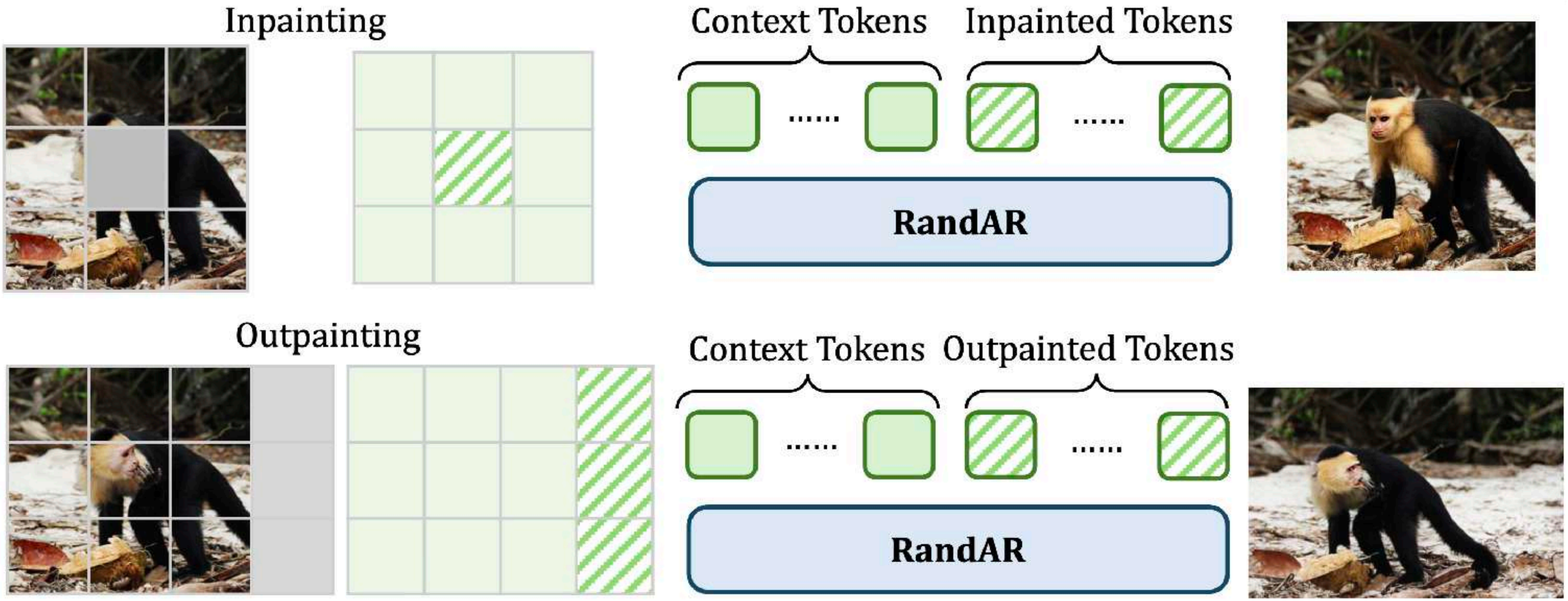
Does any order improve image quality? No, but parallel decoding



Pang et al., RandAR: Decoder-only Autoregressive Visual Generation in Random Orders. 2024

Any-order AR models

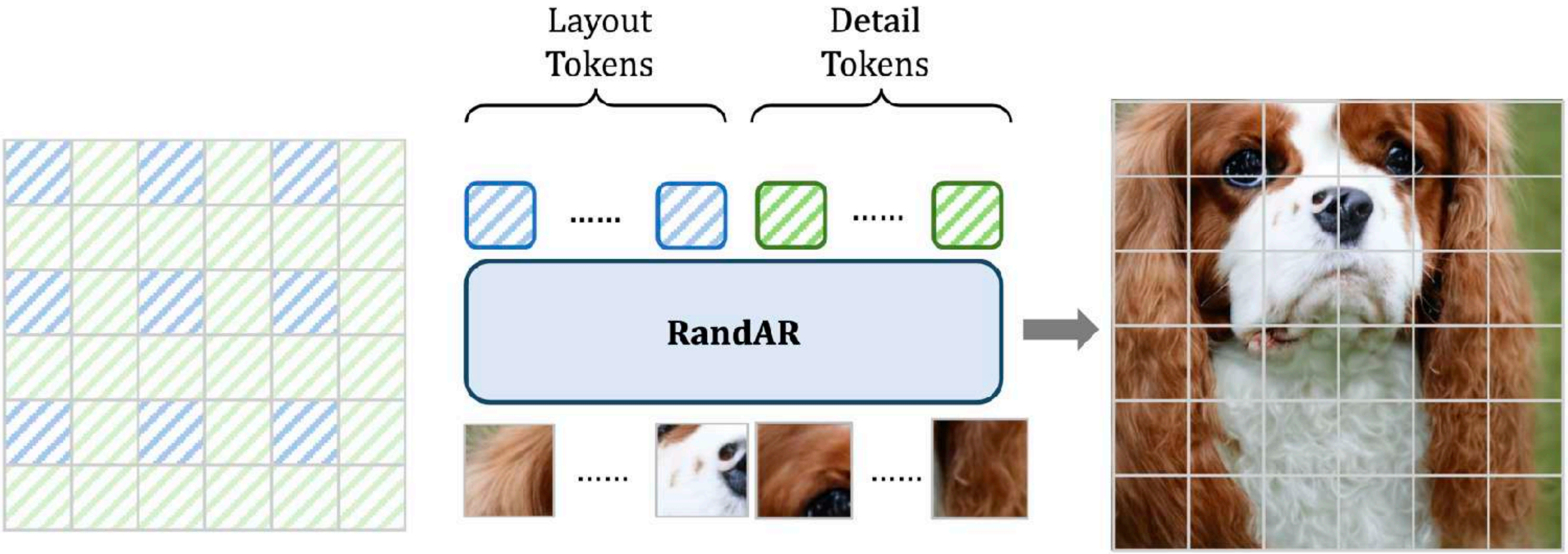
Does any order improve image quality? No, but inpainting & outpainting



Pang et al., RandAR: Decoder-only Autoregressive Visual Generation in Random Orders. 2024

Any-order AR models

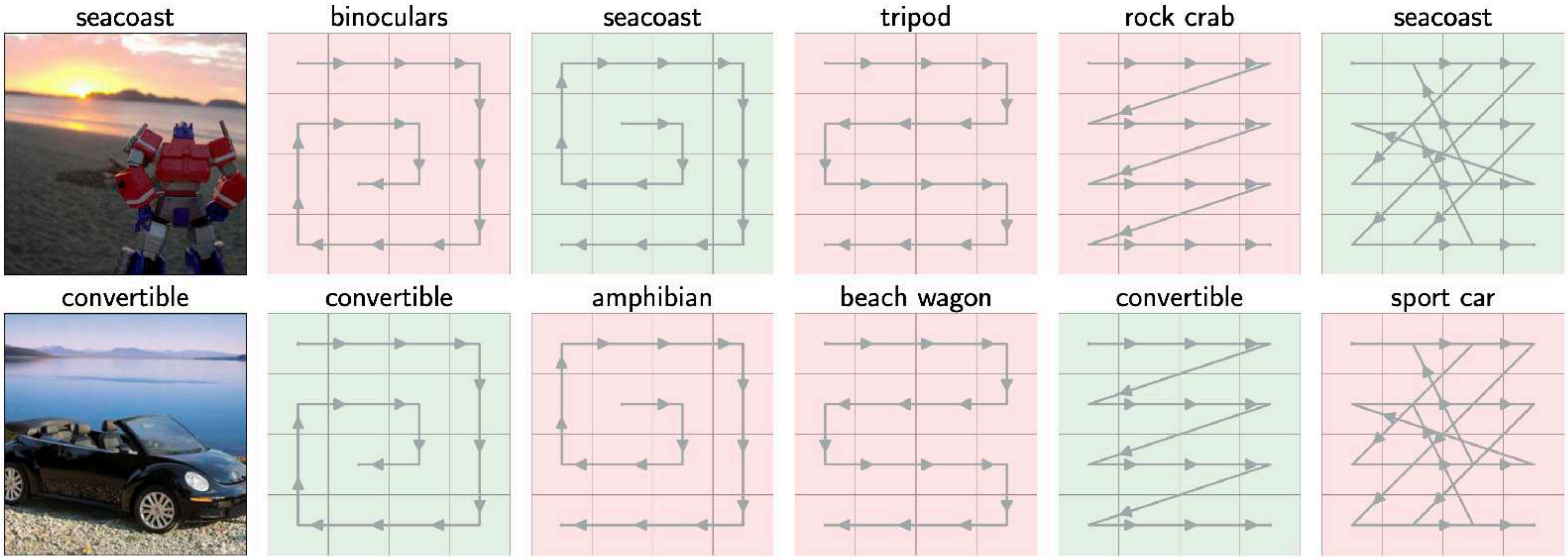
Does any order improve image quality? No, but zero-shot resolution extrapolation



Pang et al., RandAR: Decoder-only Autoregressive Visual Generation in Random Orders. 2024

Any-order AR models

Does any order improve image quality? No, but better AR-based image classification



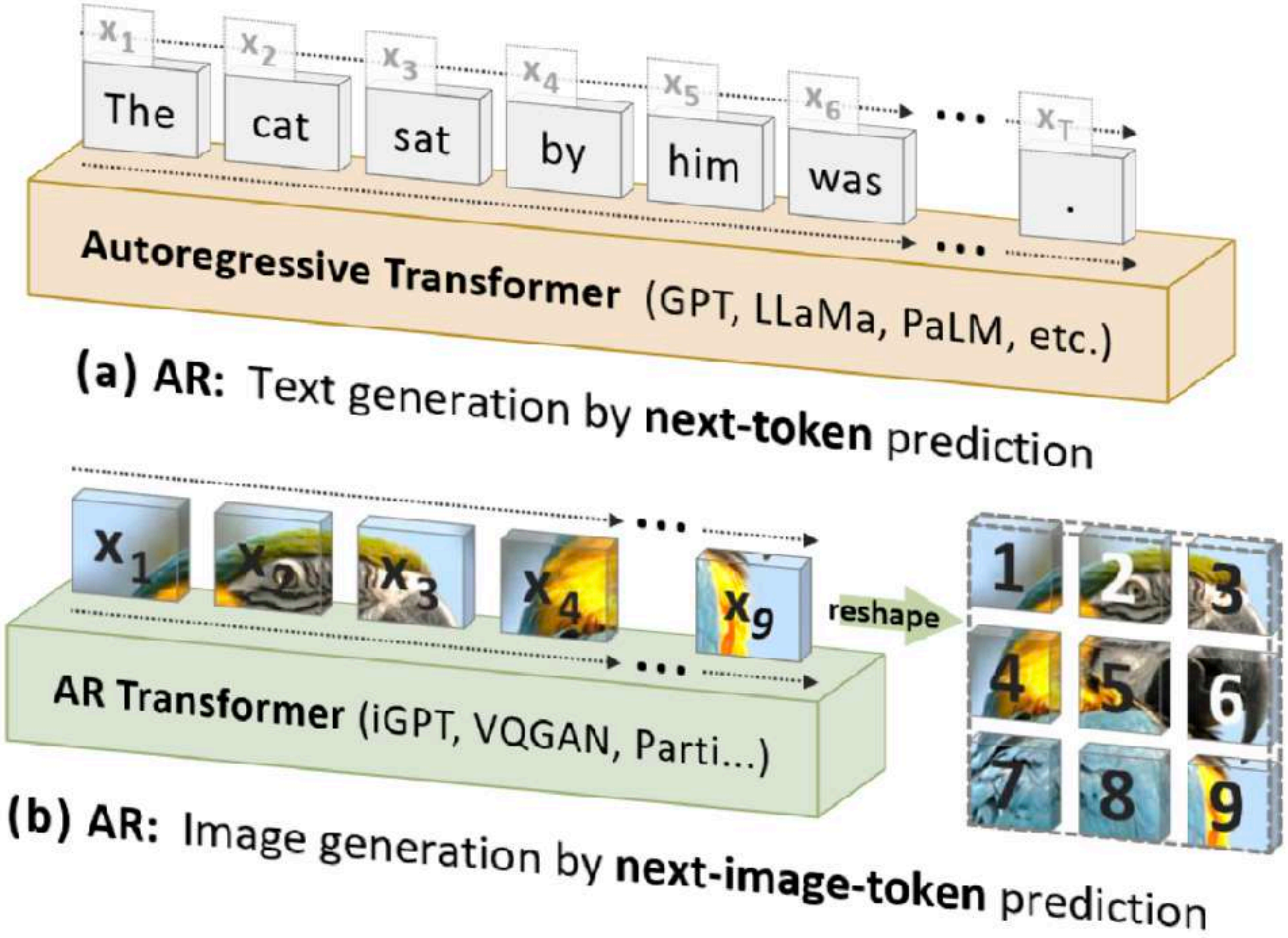
Sudakov et al., Revisiting Autoregressive Models for Generative Image Classification. 2026

Why GPT-like AR is bad for visual domains?

Next-token prediction is unnatural for visual data

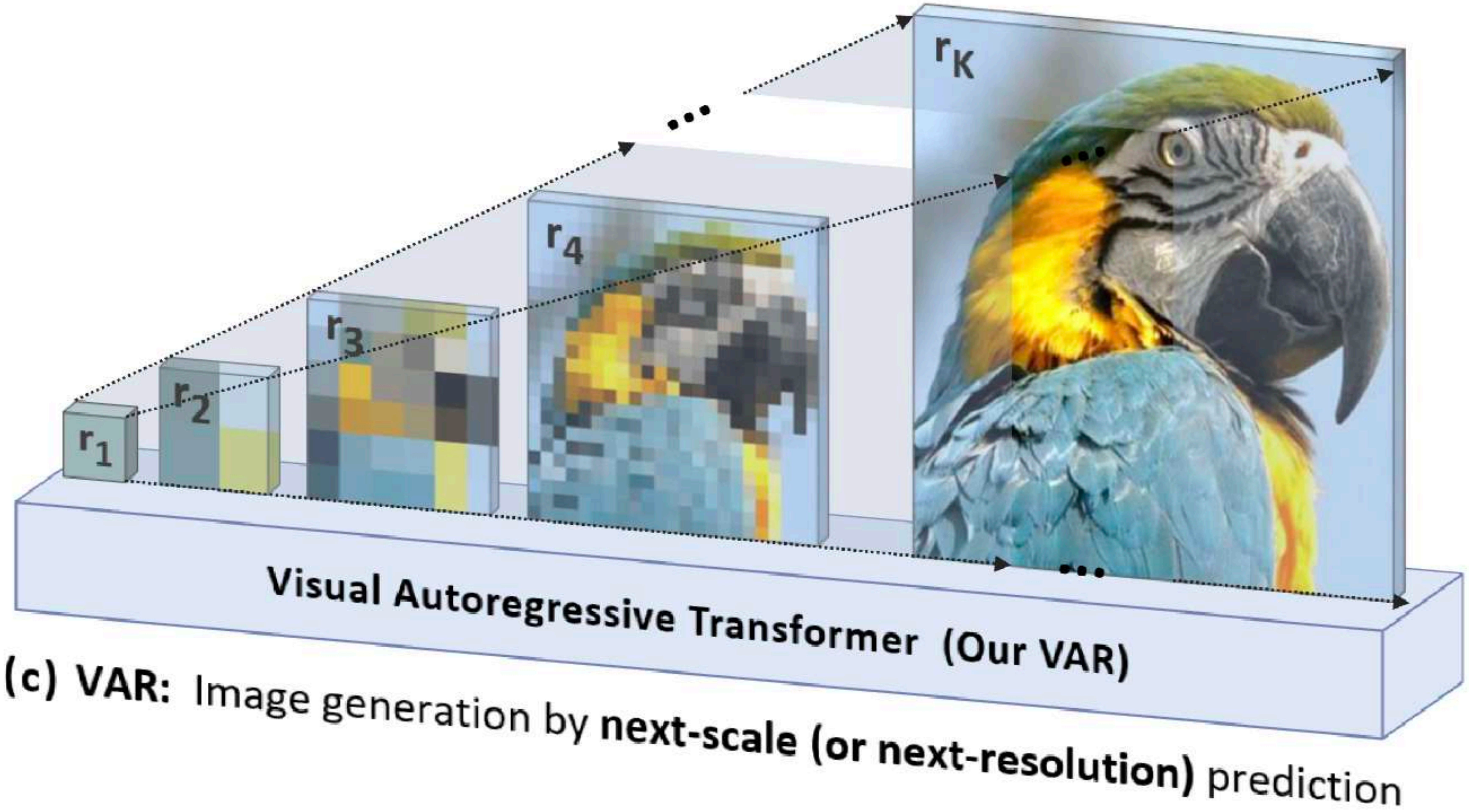
Discrete representations are worse for both image reconstruction and generators

Low inference speed even in latent space



Next-scale prediction AR models

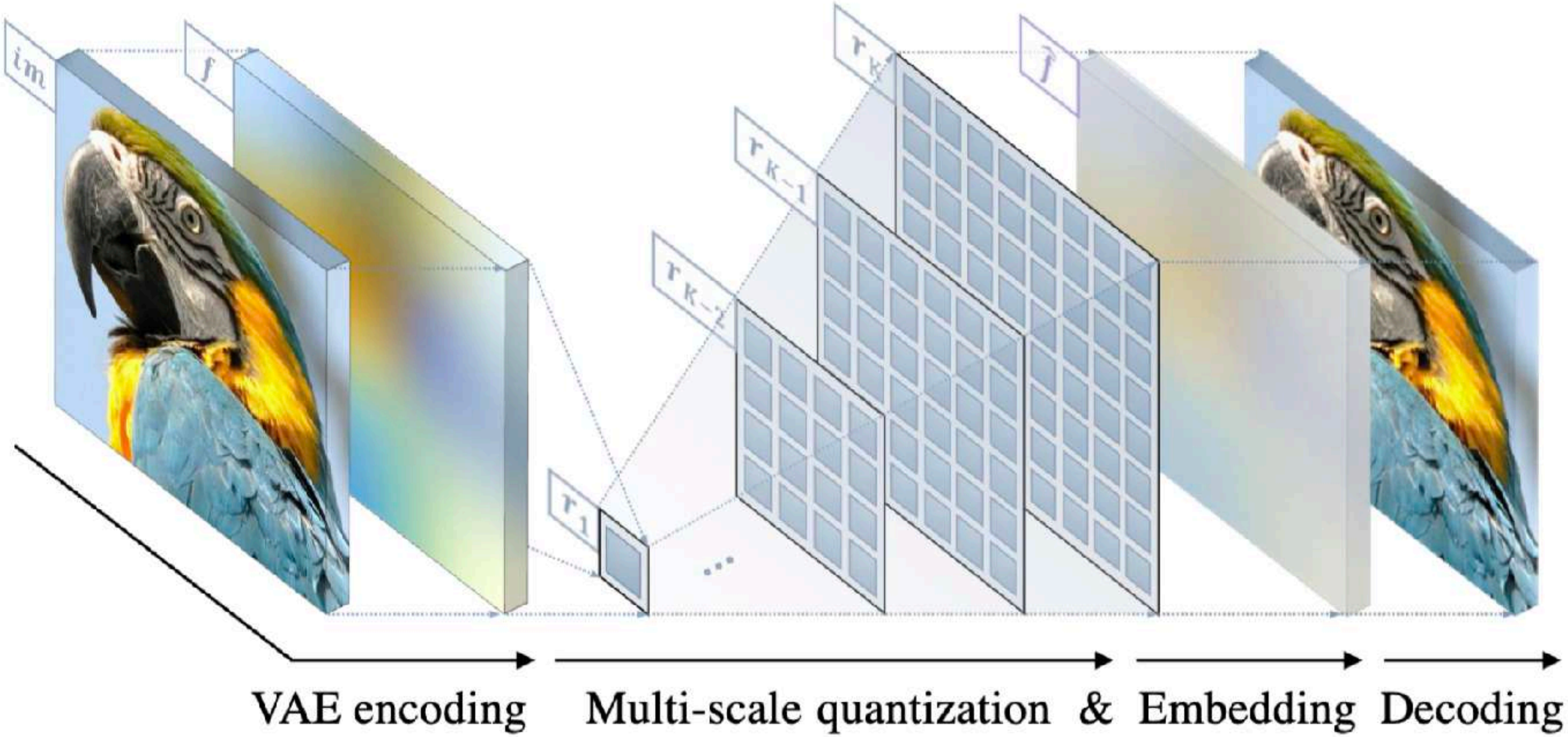
Scale-wise Visual AR Generation



Tian et al. Scalable Image Generation via Next-Scale Prediction, 2024

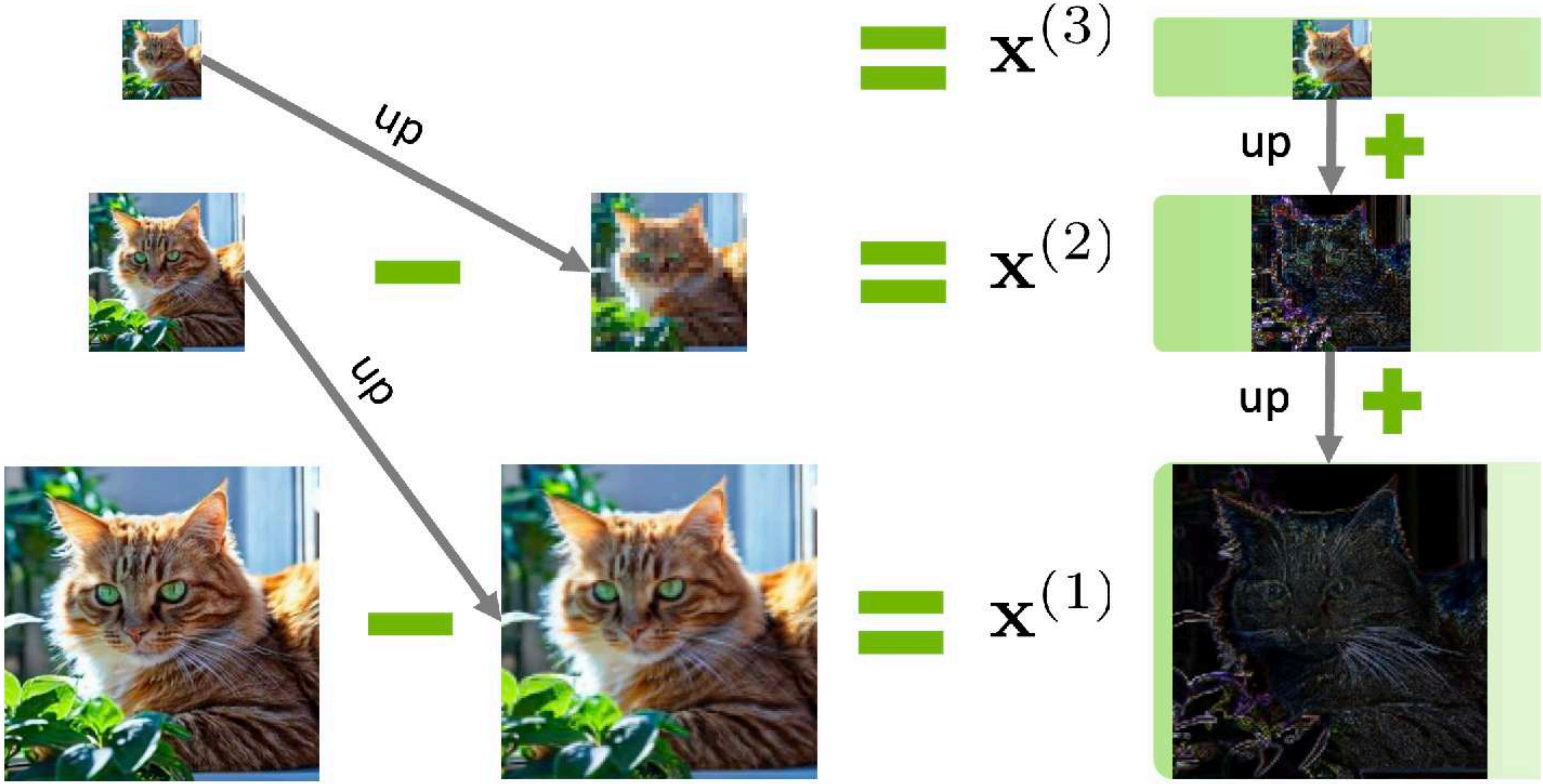
Hierarchical multi-scale VQ-VAE

Residual Quantization: $\hat{\mathbf{f}} = up(\mathbf{r}_1, 8\times) + up(\mathbf{r}_2, 4\times) + up(\mathbf{r}_3, 2\times) + \mathbf{r}_4$



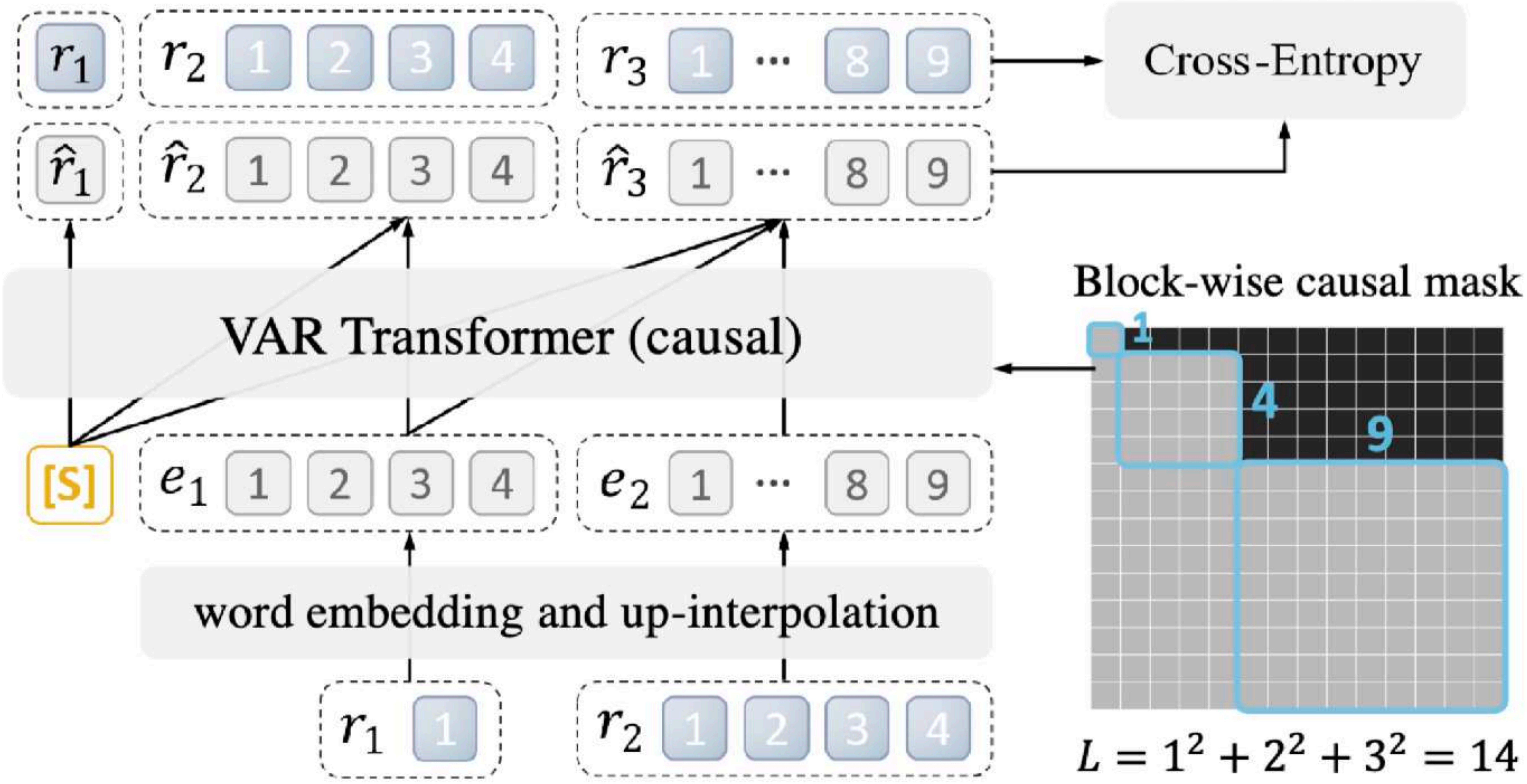
Tian et al. Scalable Image Generation via Next-Scale Prediction, 2024

Laplacian decomposition



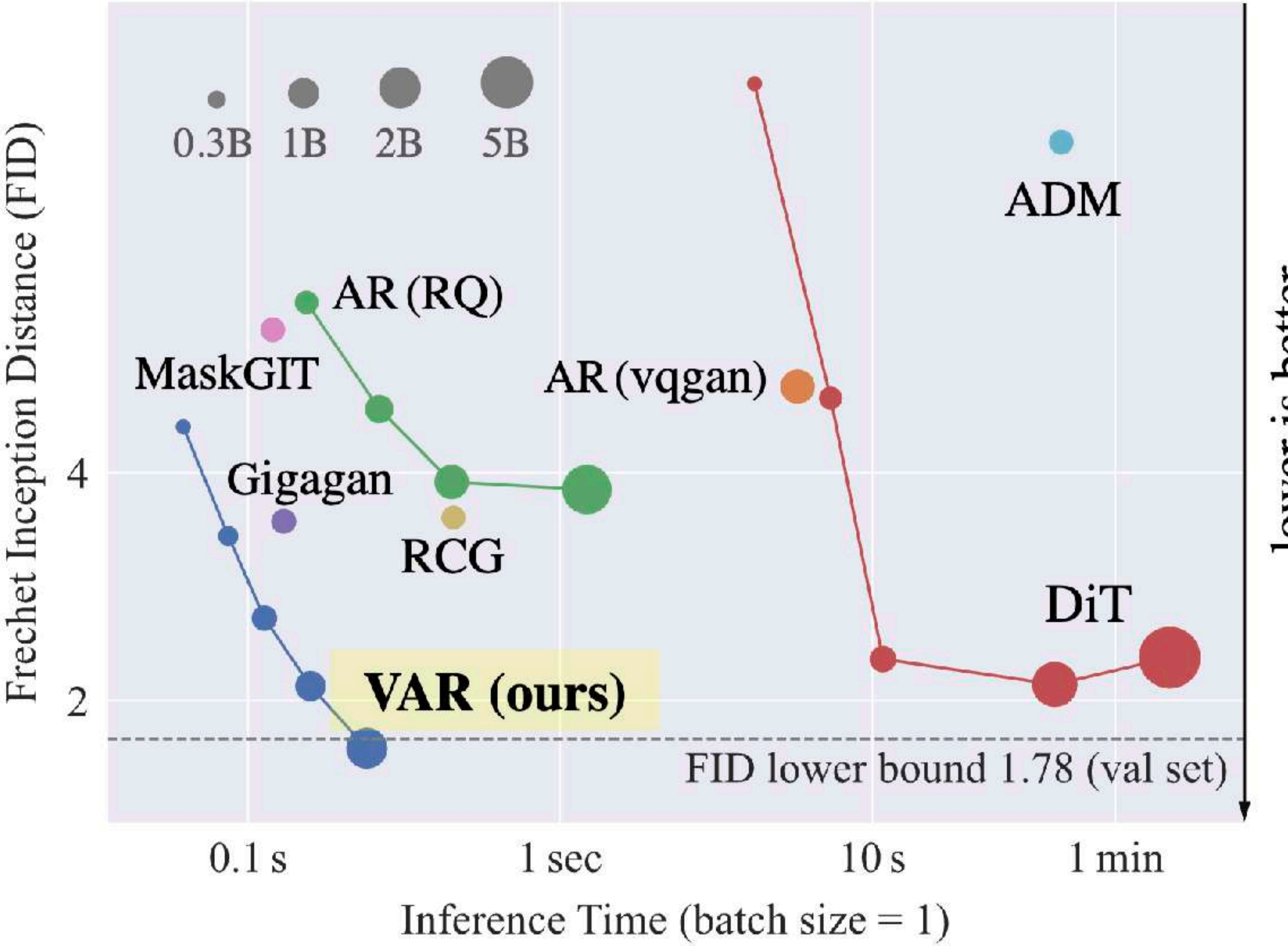
NVIDIA. Edify Image: High-Quality Image Generation with Pixel Space Laplacian Diffusion Models, 2024

VAR training



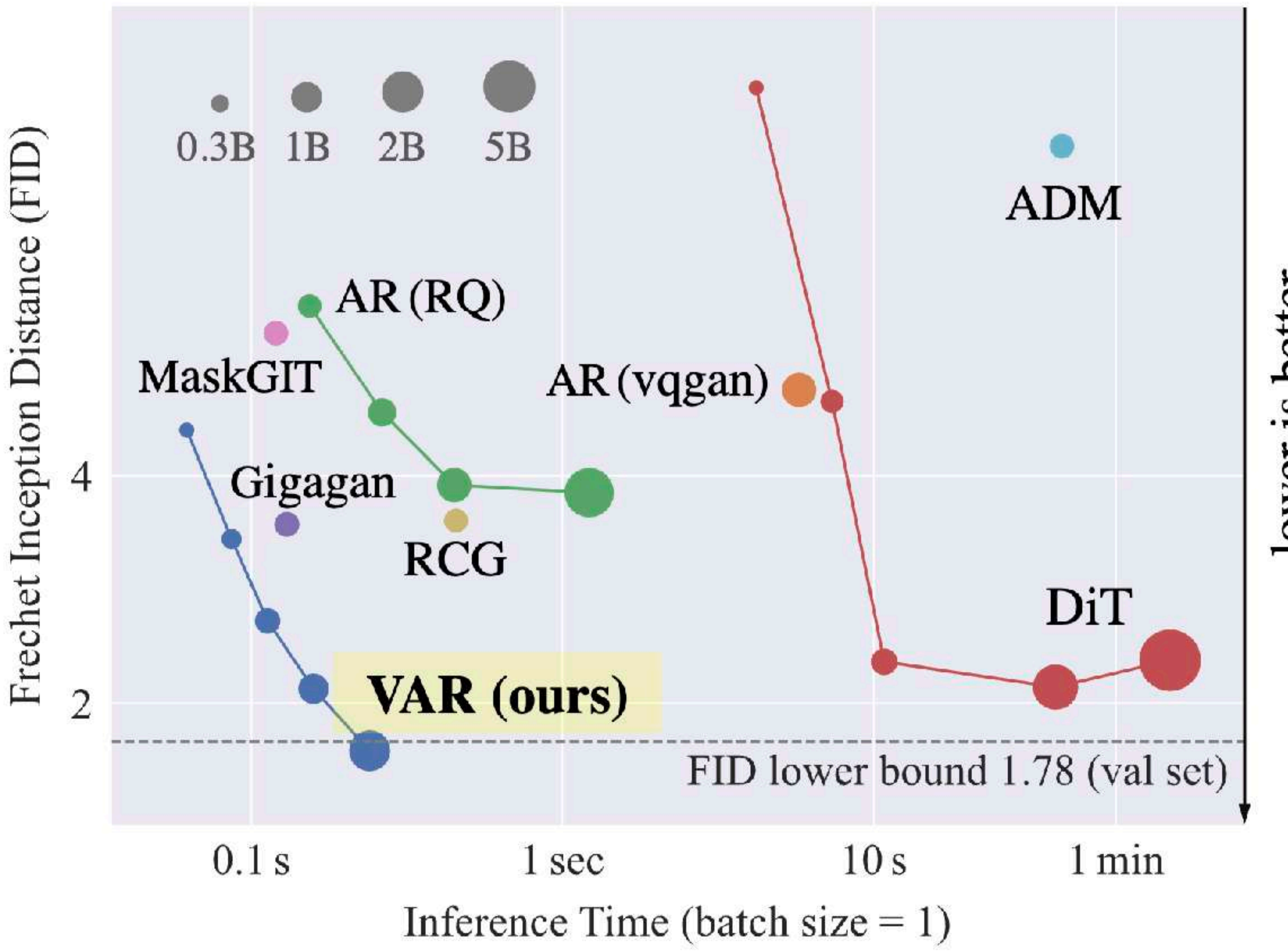
Tian et al. Scalable Image Generation via Next-Scale Prediction, 2024

VAR results


0.3B
1B
2B
5B
ADM
AR (RQ)
MaskGIT
AR (vqgan)
Gigagan
RCG
DiT
VAR (ours)
FID lower bound 1.78 (val set)
Frechet Inception Distance (FID)
4
2
0.1 s
1 sec
10 s
1 min
Inference Time (batch size = 1)

VAR results

What about text-to-image generation?



SWITTI: Designing Scale-Wise Transformers for Text-to-Image Synthesis

Anton Voronov[1,2,3] Denis Kuznedelev[1,4] Mikhail Khoroshikh[6] Valentin Khrulkov[1,5]
Dmitry Baranchuk[1]

[1]Yandex Research [2]HSE University [3]MIPT [4]Skoltech [5]AIRI [6]ITMO

https://yandex-research.github.io/switti

Figure 1. SWITTI produces high quality and aesthetic 1024×1024 image samples in around 0.5 seconds.

Designing and Scaling T2I VAR

Goal: design and train first T2I VAR

Project directions

T2I VAR Architecture

CLIP text conditioning via CA and AdaLN

Non-causal attention

"Sandwich" normalization for stability

Multi-scale RoPE

Large-scale training

2.5B model

100M dataset

320 A100 GPUs

256 → 512 → 1024 stages

SFT post-training

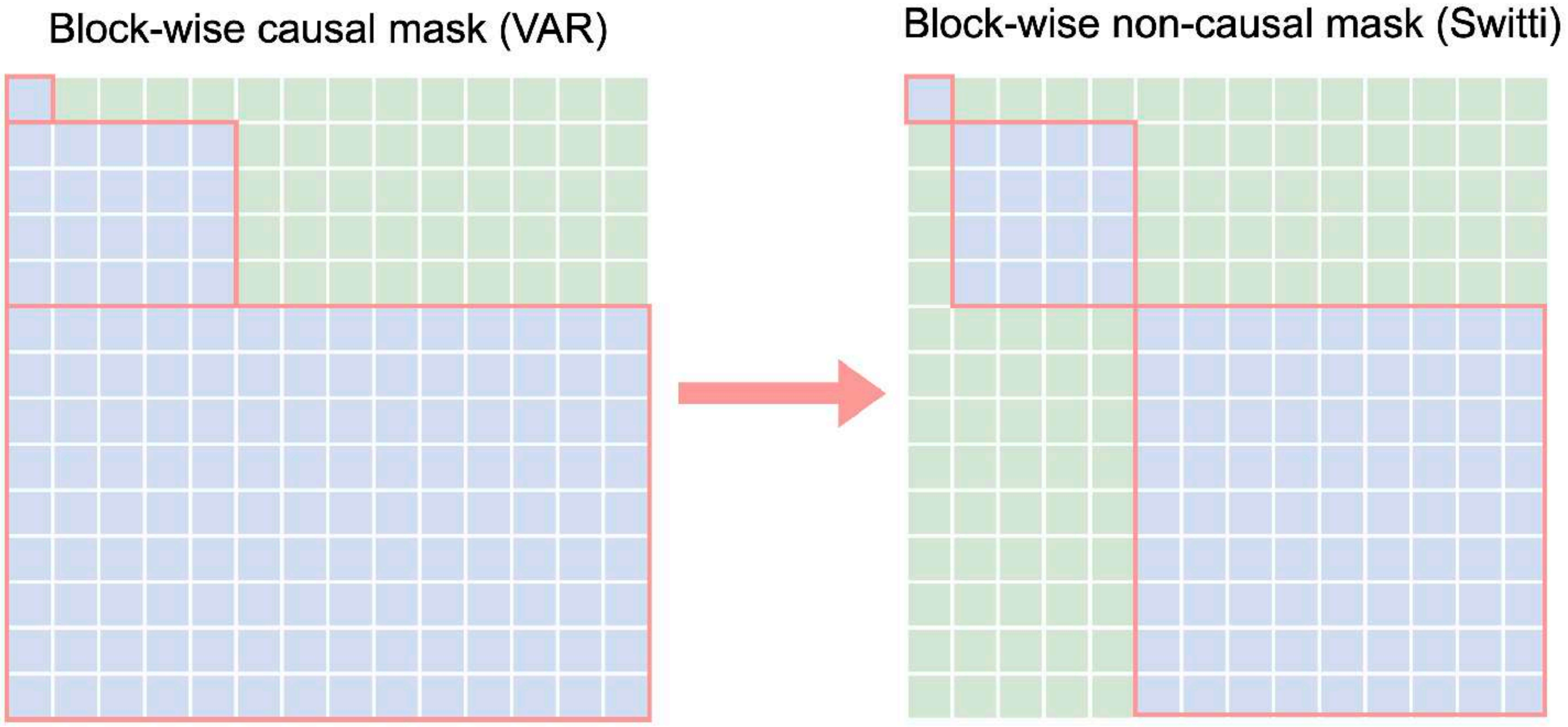
Analysis

Activation norms

Attention behavior

Role of text-conditioning

Switti is no longer explicitly autoregressive



Switti is implicitly autoregressive

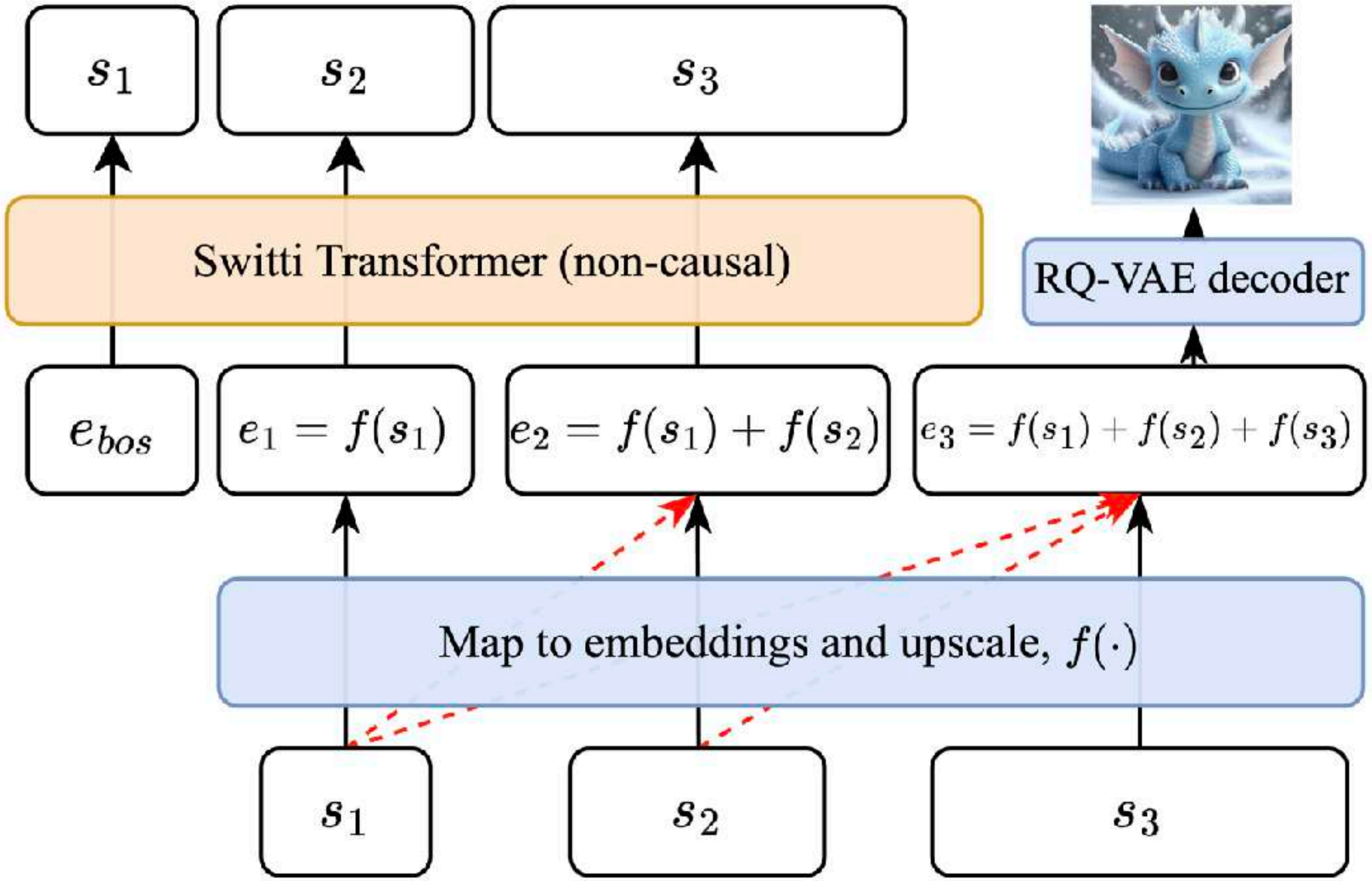
Flexible training

- No need to process all scales at once

More efficient inference

- No KV-cache (saves 2.4GB/image)
- Cheaper attention calculation (~21%)

Slightly higher quality



Effect of text conditioning

Weak text reliance at last scales

↓

Turn off CFG

~32% speedup

Less defects

VAR Summary

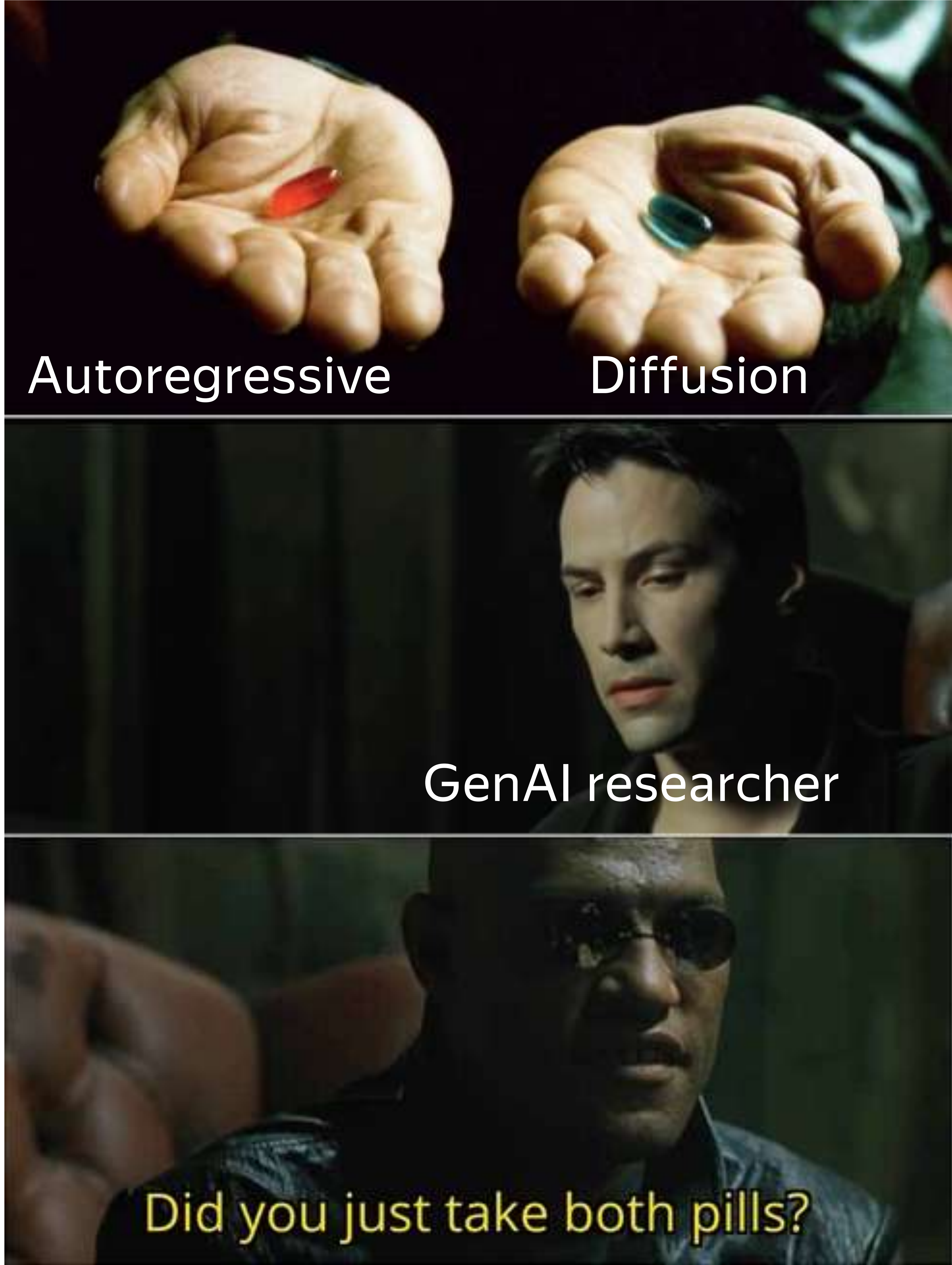
Highly natural visual inductive bias

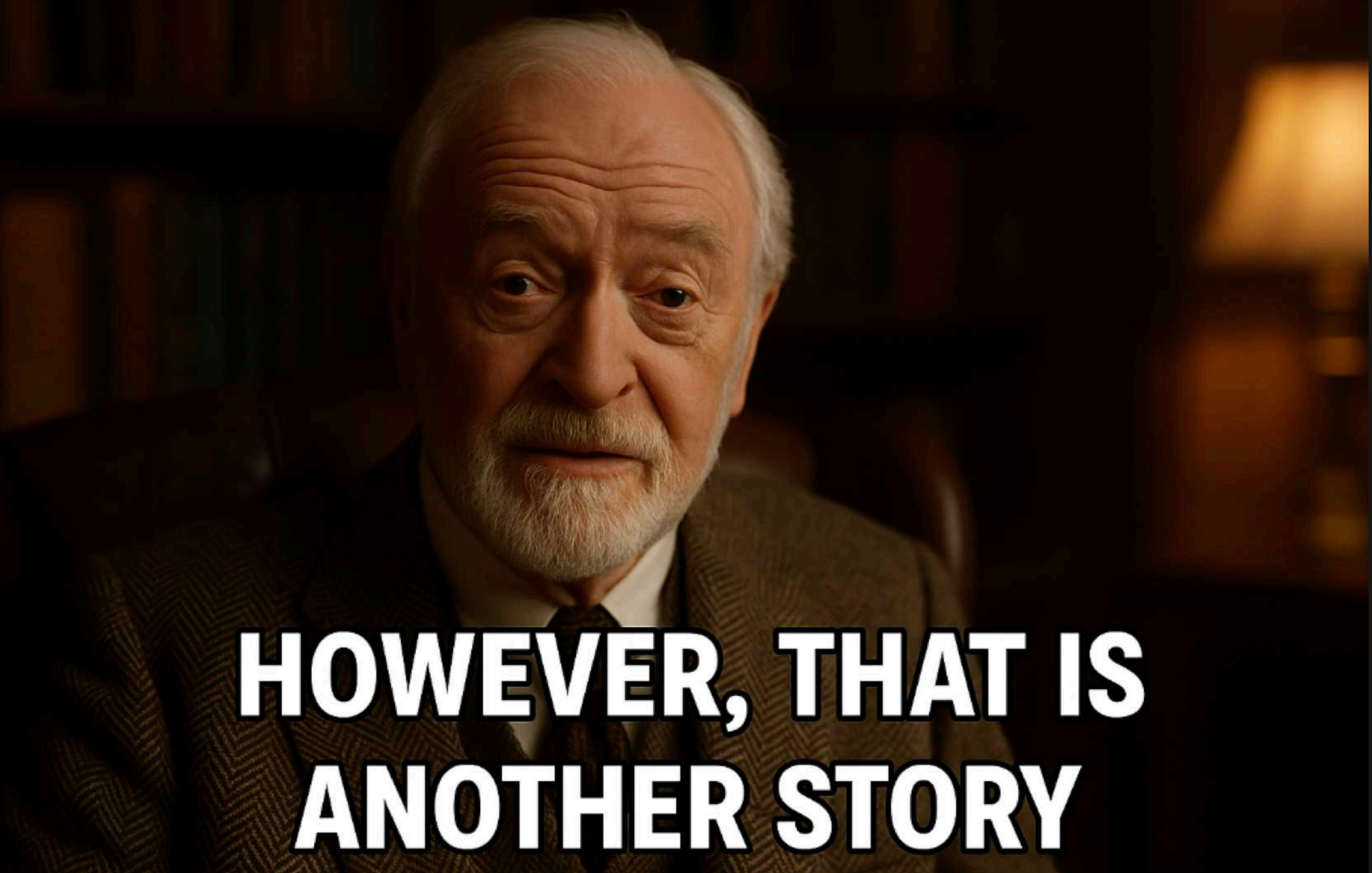
Much faster than DMs by design

Still requires discrete tokenizers

Promising results but still needs much work to achieve top-tier DM performance

Continuous AR models

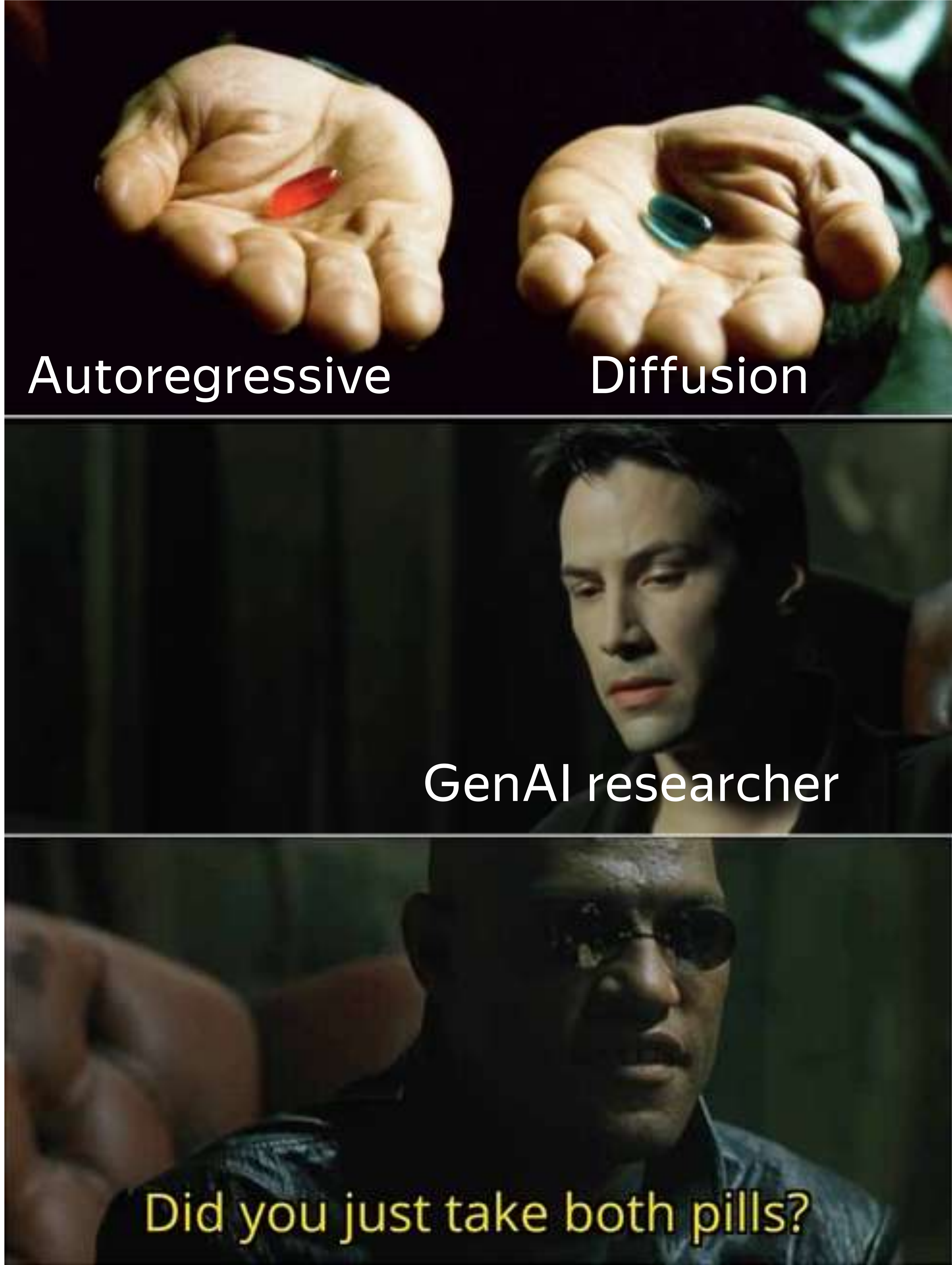



HOWEVER, THAT IS
ANOTHER STORY

И сгенерируй теперь прощальную картинку где он машет рукой

Изображение создано

Continuous-token AR models



Why discrete tokenizers?

Discrete AR models

For AR generation, we need to define $p(\mathbf{x} \mid \mathbf{z})$
$\mathbf{x}$ — discrete VQ token
$\mathbf{z}$ — predicted *continuous* token presentation prior to quantization

$p(\mathbf{x} \mid \mathbf{z})$ — categorical distribution: $softmax(W\mathbf{z}/\tau)$, where W is a codebook

Loss: cross-entropy

Sampling: Gumbel-Max or inverse transform sampling

What about continuous tokenizers?

Continuous AR models

For AR generation, we need to define $p(\mathbf{x} \mid \mathbf{z})$

$\mathbf{x}$ — *continuous* token, e.g., a latent patch of 16×16

$\mathbf{z}$ — predicted *continuous* patch representation with the AR model

$p(\mathbf{x} \mid \mathbf{z}) - ?$

Loss: $?$

Sampling: $?$

What about continuous tokenizers?

Continuous AR models

For AR generation, we need to define $p(\mathbf{x} \,|\, \mathbf{z})$

$\mathbf{x}$ — *continuous* image/latent patch, e.g., 16×16

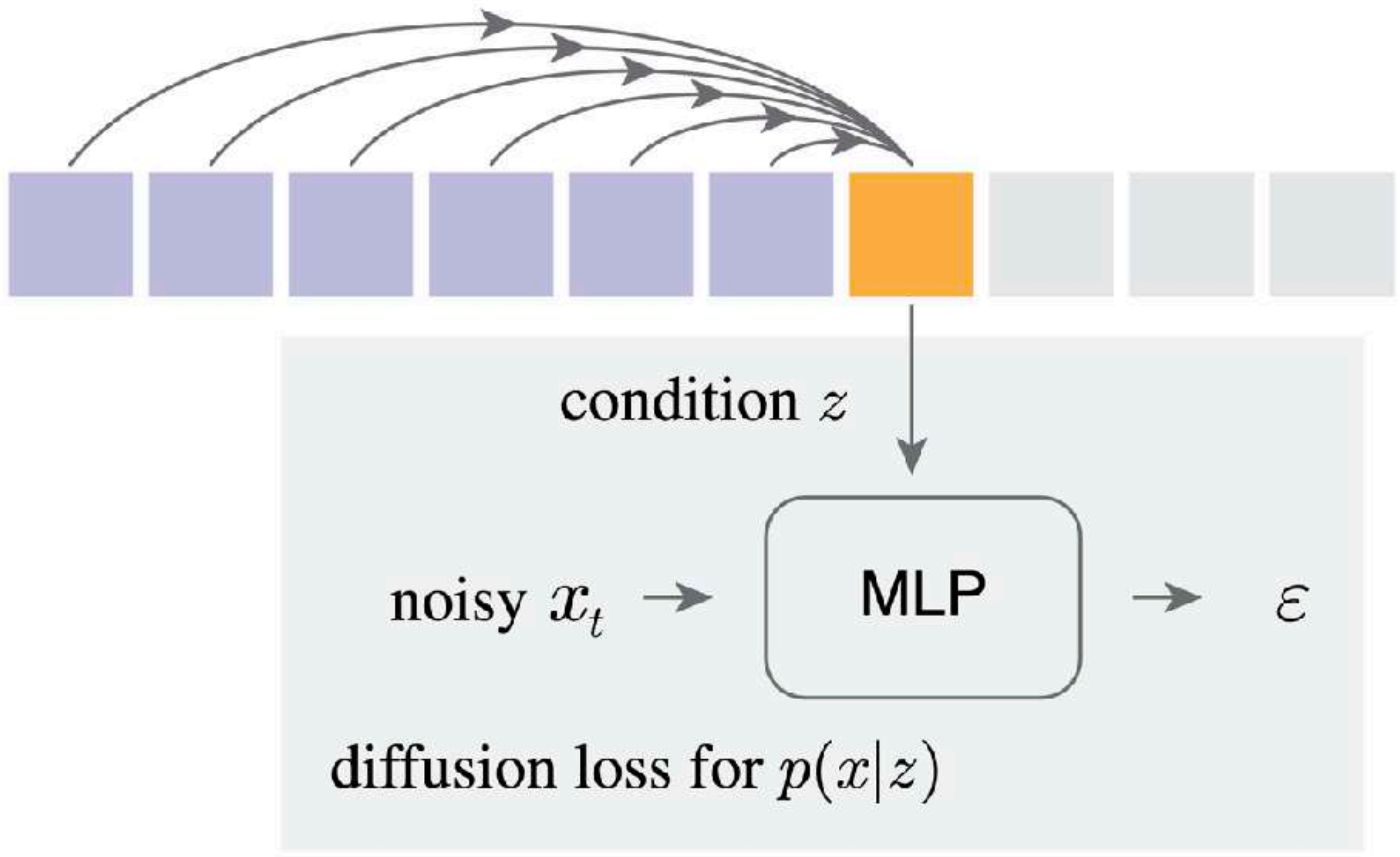
$\mathbf{z}$ — predicted *continuous* patch representation with the AR model

$p(\mathbf{x} \,|\, \mathbf{z})$ — **diffusion**

Loss: diffusion

Sampling: diffusion

Diffusion loss



Diffusion loss plays a role of perceptual or adversarial losses

Diffusion loss

$$\mathcal{L}(z, x) = \mathbb{E}_{\varepsilon,t}\left[\left\|\varepsilon - \varepsilon_\theta(x_t|t, z)\right\|^2\right]$$

Li et al. Autoregressive Image Generation without Vector Quantization, 2024

Denoising MLP

3-layer MLP model | 2M parameters — very small compared to the AR model

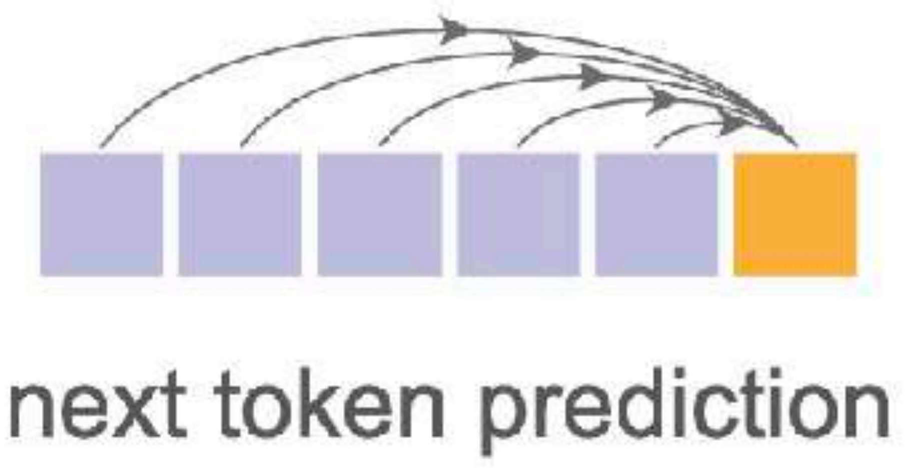
100 steps is sufficient and takes minor compute overhead at inference (~10%)

Temperature (CFG) is important, similar to the discrete case

Li et al. Autoregressive Image Generation without Vector Quantization, 2024

Getting back to prediction strategies

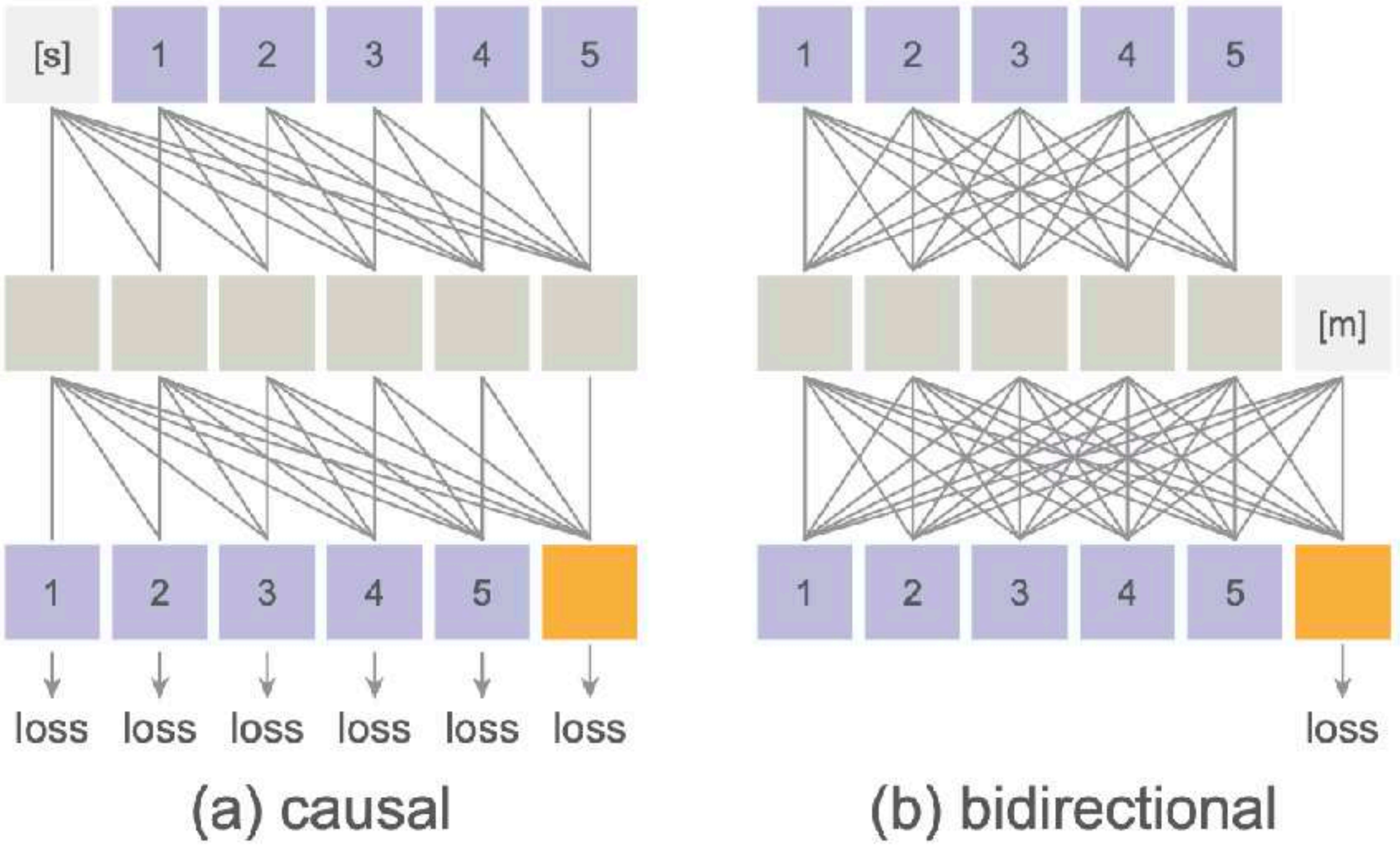
Next-token prediction can be done with bidirectional transformers



Bidirectional models

Allow multiple predictions at each step

Do not allow using KV-cache



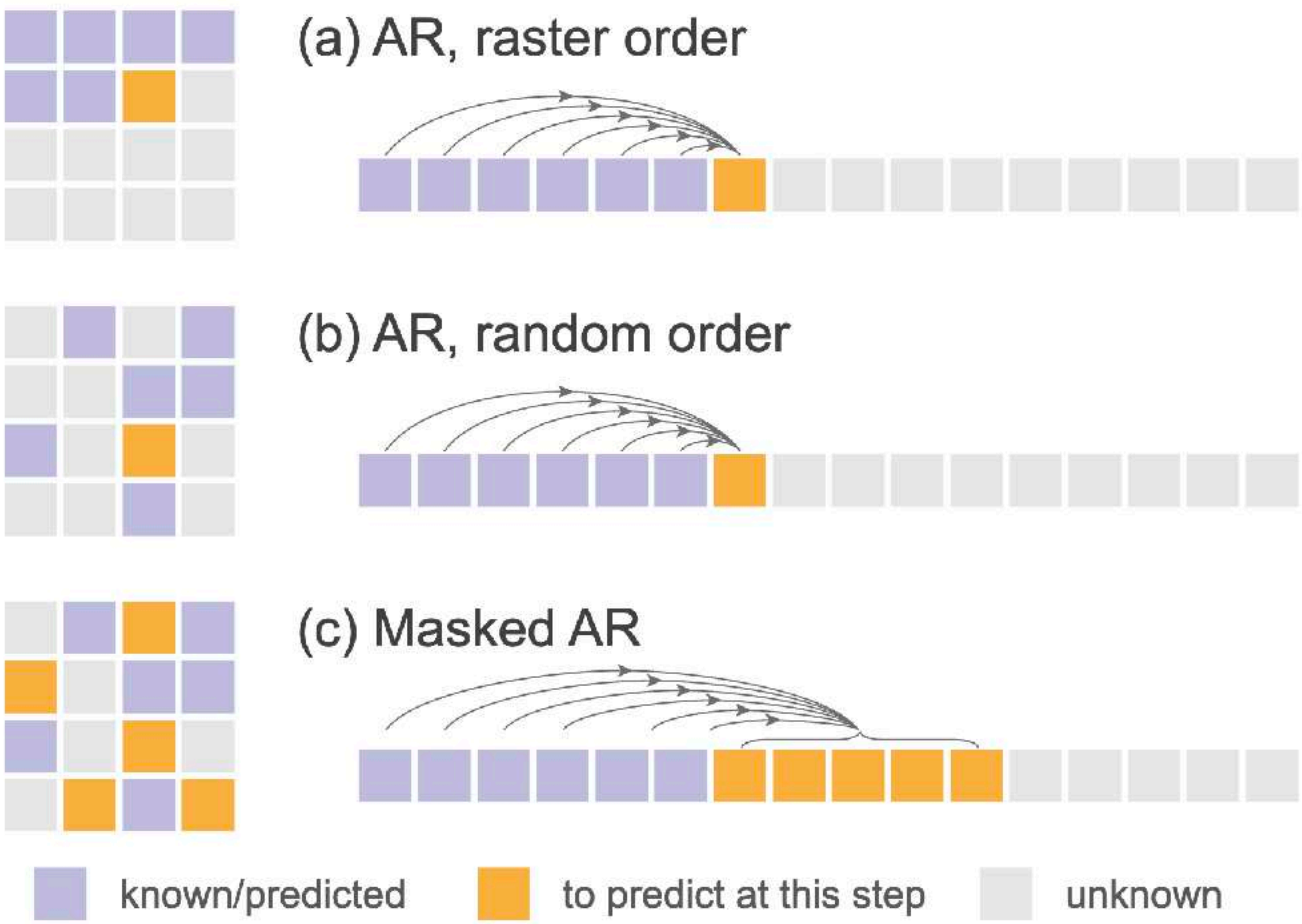
Li et al. Autoregressive Image Generation without Vector Quantization, 2024

Masked autoregressive models

Masked AR is a form of the
next-set prediction strategy

In fact, only predicted tokens
need a bidirectional mask

Promising variant
Block-wise causal like VAR
not discussed in the paper



Li et al. Autoregressive Image Generation without Vector Quantization, 2024

Ablation

Random-order > raster-order

Bidirectional > causal

Continuous > discrete: VAE > VQ-VAE and diffusion loss > cross-entropy

variant	order	direction	# preds	loss	w/o CFG FID↓	w/o CFG IS↑	w/ CFG FID↓	w/ CFG IS↑
AR	raster	causal	1	CrossEnt	19.58	60.8	4.92	227.3
				Diff Loss	**19.23**	**62.3**	**4.69**	**244.6**
MAR	rand	causal	1	CrossEnt	16.22	81.3	4.36	222.7
				Diff Loss	**13.07**	**91.4**	**4.07**	**232.4**
MAR	rand	bidirect	1	CrossEnt	8.75	149.6	3.50	280.9
				Diff Loss	**3.43**	**203.1**	**1.84**	**292.7**
MAR (default)	rand	bidirect	>1	CrossEnt	8.79	146.1	3.69	278.4
				Diff Loss	**3.50**	**201.4**	**1.98**	**290.3**

Li et al. Autoregressive Image Generation without Vector Quantization, 2024

Text-to-image MAR | Fluid

FLUID: SCALING AUTOREGRESSIVE TEXT-TO-IMAGE GENERATIVE MODELS WITH CONTINUOUS TOKENS

Lijie Fan[1,*] **Tianhong Li**[2,†] **Siyang Qin**[1,†] **Yuanzhen Li**[1] **Chen Sun**[1]
Michael Rubinstein[1] **Deqing Sun**[1] **Kaiming He**[2] **Yonglong Tian**[1,*]
[1]Google DeepMind [2]MIT [*] equal contribution, project lead [†] equal contribution

Fan et al. Fluid: Scaling Autoregressive Text-to-image Generative Models with Continuous Tokens. 2024

Multimodal MAR | UniFluid

Unified Autoregressive Visual Generation and Understanding with Continuous Tokens

Lijie Fan[1,*] Luming Tang[1,*] Siyang Qin[1,*] Tianhong Li[2] Xuan Yang[1] Siyuan Qiao[1]
Andreas Steiner[1] Chen Sun[1] Yuanzhen Li[1] Tao Zhu[1] Michael Rubinstein[1] Michalis Raptis[1]
Deqing Sun[1,†] Radu Soricut[1,†]
[1]Google DeepMind [2]MIT [*,†] equal contribution

Fan et al. Unified Autoregressive Visual Generation and Understanding with Continuous Tokens. 2025

MAR Summary

Eliminating discreteness in visual AR models via diffusion loss

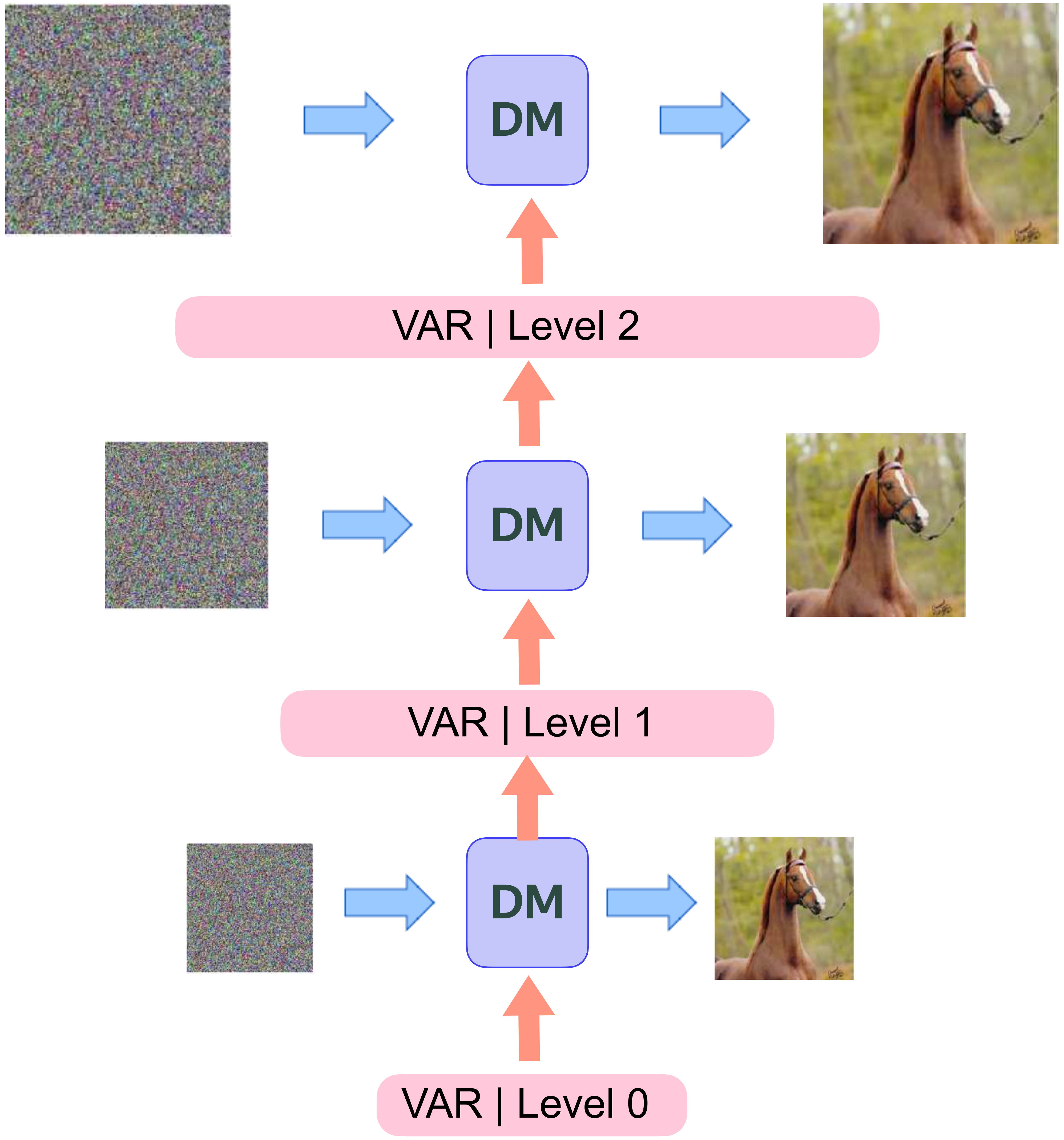
Prediction strategy seems suboptimal but well explored in other works

What about ...

MAR + VAR

Reasonable but not quite elegant…
Looks like a cascaded DM

Do we know any other scale-wise and continuous AR paradigms?



Ten et al. Next-X Prediction for Autoregressive Visual Generation. 2025

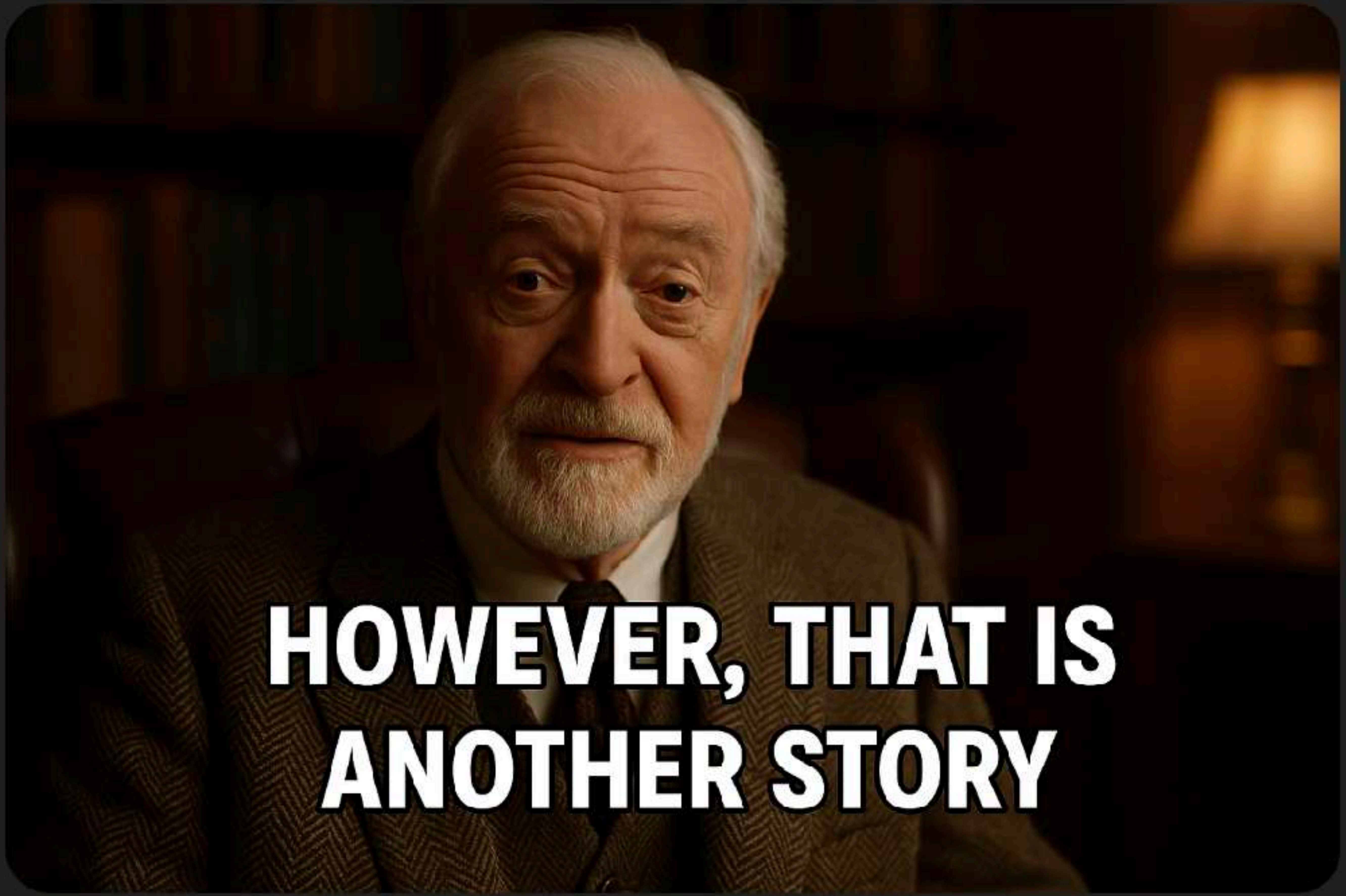
Diffusion models

Implicit AR models

Implicit scale-wise generation

Сгенериурй мем "Впрочем, это уже совсем другая история", только сделай надпись на английском

Изображение создано



И сгенерируй теперь прощальную картинку где он машет рукой

Изображение создано